

# SpaceLogic C-Bus Digital Dimmers

## User Guide

Information about features and functionality of the device

5508D1D

5504D2D

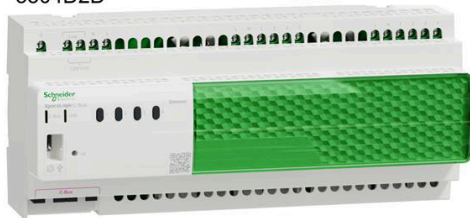
5504DHD

Release Date 05/2025

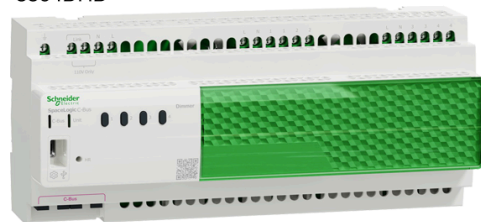
5508D1D



5504D2D



5504DHD



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# Safety Information

## Important Information

Read these instructions carefully and observe the equipment to become familiar with the device before trying to install, operate, service, or maintain it. The following special messages may appear throughout this manual or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of a symbol to either a “Danger” or “Warning” safety label indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that accompany this symbol to avoid possible injury or death.

### **DANGER**

**DANGER** indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

**Failure to follow these instructions will result in death or serious injury.**

### **WARNING**

**WARNING** indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

### **CAUTION**

**CAUTION** indicates a hazardous situation which, if not avoided, **could result in** minor or moderate injury.

### **NOTICE**

**NOTICE** is used to address practices not related to physical injury.

## Note

Electrical equipment must be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction, installation, and operation of electrical equipment and has received safety training to recognize and avoid the hazards involved.

## Safety Precautions

### **⚡⚠ DANGER**

#### **HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH**

- This product must be installed and serviced by appropriately qualified and/or licensed professional in accordance with the local wiring rules.
- Isolate the electrical supply before doing any work on the product.
- Ensure that the product has been correctly installed and tested for safe operation before reconnecting the electrical supply.
- Do not use this product for any other purpose than specified in this instruction.
- Pay attention to the specifications and wiring diagrams related to the installation.
- Do not attempt to open the product casing or perform any action on the internal components of the product.

**Failure to follow these instructions will result in death or serious injury.**

### **⚠ WARNING**

#### **INCORRECT C-BUS CABLE INSTALLATION**

The C-Bus network cabling is classified as Separated Extra-Low Voltage (SELV) wiring. To maintain this requirement, the approved C-Bus cable must be used.

- Ensure that adequate separation and/or segregation of the C-Bus cable from other wiring (for example Low Voltage wiring) is maintained throughout the entire installation.
- Ensure the C-Bus network cable is installed in accordance with the SELV wiring rules and local safety standards.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

### **NOTICE**

#### **INCORRECT C-BUS INSTALLATION LOCATION**

To prevent the possibility of intentional or unintentional interference with the configuration or operation of the C-Bus installation, this equipment should be installed in a location with appropriate access control.

**Failure to follow these instructions can result in equipment damage.**

### **NOTICE**

#### **EQUIPMENT DAMAGE**

- Use of non-dimmable loads (including non-dimmable loads connected the secondary of an iron-core transformer) can reduce the effectiveness of the dimmer's inbuilt protection mechanisms. Operation of these types of load may result in permanent damage to the dimmer channel.
- Never connect a load to a channel whilst the channel is powered this includes the changing lamp fittings. Doing so reduces the ability of the dimmer channel to protect itself against incompatible loads and doing so with some types of load may result in permanent damage to the dimmer channel.

**Failure to follow these instructions can result in equipment damage.**

## Megger Testing

### ***NOTICE***

#### **MEGGER TESTING**

Megger testing must never be performed on any cable while connected to the product as it could degrade the performance of the product and/or the network.

**Failure to follow these instructions can result in equipment damage.**

### ***NOTICE***

#### **MAXIMUM LOAD RATINGS APPLY**

Ensure that the number of lighting loads connected to a single device does not exceed the maximum load rating of each channel.

**Failure to follow these instructions can result in equipment damage.**

## Cybersecurity Information

At Schneider Electric, we believe that Cybersecurity is an essential prerequisite. We are committed to providing reliable, stable, and secure products to minimize potential network risks and protect the safety of customers, property, and the environment.

Cybersecurity aims to prevent your system, communication networks, and devices from possible attacks, data tampering, or confidential information leakage.

In addition to the direct instructions in this document, observe and follow Schneider Electric's security recommendations. For details and assistance in protecting your installation, you can also contact your local Schneider Electric Industrial Cybersecurity Services organization or visit Cybersecurity Services on the Schneider Electric website.

**The below table describes the Cybersecurity Services on the Schneider Electric Website. For more information, click on the link.**

| Link                                     | Services                                                                                                        |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Recommended Cybersecurity Best Practices | Proven Cybersecurity procedures.                                                                                |
| Cyber security service                   | From conception to maintenance: certified experts advise to guide you through a holistic Cybersecurity program. |
| Cybersecurity support portal             | Security notifications, reporting a vulnerability, reporting an incident.                                       |

## Cybersecurity Vulnerabilities/Incidents

You can review the Vulnerability Management Policies on Schneider Electric's Cybersecurity Vulnerabilities Portal (<https://www.se.com/ww/en/work/support/cybersecurity/vulnerability-policy.jsp>) or report potential Cybersecurity vulnerabilities or incidents.

## General Cybersecurity Information

In recent years, the growing number of networked machines and production plants has seen a corresponding increase in the potential for cyber threats, such as unauthorized access, data breaches, and operational disruptions. You must, therefore, consider all possible cybersecurity measures to help protect assets and systems against such threats.

To help keep your Schneider Electric products secure and protected, it is in your best interest to implement the cybersecurity best practices as described in the [Cybersecurity Best Practices](#) document.

Schneider Electric provides additional information and assistance:

- [Subscribe to the Schneider Electric security newsletter.](#)
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The product must be installed and used in accordance with all applicable state, regional, and local safety regulations. In order to ensure safety and compliance with documented system data, only the manufacturer should perform component repairs.

Devices with technical safety requirements must follow the relevant instructions.

Failure to use Schneider Electric software or approved software with our hardware products may result in injury, harm, or improper operating results.

Failure to observe this information can result in injury or equipment damage.  
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## About the Device

The 5508D1D, 5504D2D and 5504DHD are new generation C-Bus phase cut dimmers compatible with a wide range of load types due to their customizable load profile and dimming curve features, designed for switchboard distribution applications.

The units are DIN rail mounted, measuring 12 modules wide (1 module = 18 mm).

C-Bus connection is achieved through the use of RJ45 connectors, allowing similar units to be looped together quickly.

Below listed are the specifications of dimmers:

- DIN Rail mount, 12 M size
- Integrated Switchable C-Bus Power Supply
- Trailing-edge or Leading-edge phase control (selectable per channel)
- Fully customizable load profiles and dimming curves
- Suitable for 230V ~ 50Hz or 110V ~ 60Hz nominal supply voltage
- Advanced status information for the unit and the channels
- Two phase-independent banks (5508D1D and 5504D2D)
- Phase Independent Channels (5504DHD only)
- Upgradeable firmware
- Full configuration with SpaceLogic C-Bus Commission
- Programming via C-Bus connection without an energized mains line
- 3 models
  - 8 channel 1 A (5508D1D)
  - 4 channel 2 A (5504D2D)
  - 4 Channel High Power (5504DHD)

## Phase-Cut Dimming

A phase cut (also known as phase control) dimmer adjusts the brightness of the connected load by varying the portion of the mains AC waveform that is passed through the connected load.

In a 50 Hz system, a full mains cycle takes 20 milliseconds (ms), which consists of a 10 ms positive half-cycle and a 10 ms negative half-cycle.

At the beginning of the cycle, the mains voltage is zero (*zero crossing*), it then rises to reach its peak value (~340 V peak in a 240 V rms system) after 5 ms, before reducing again to zero at 10 ms. The cycle then repeats with the opposite polarity.

## Conduction Time

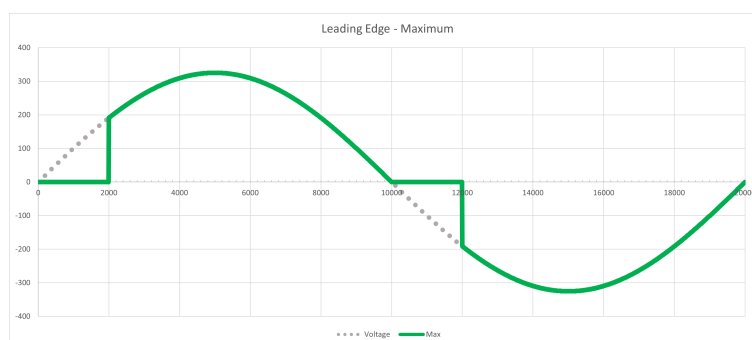
The portion of the 10 ms half-cycle time for which the dimmer passes power to the load is called the conduction time, and might be expressed in milliseconds (ms) or microseconds ( $\mu$ s) where 1 ms = 1000  $\mu$ s.

## Leading Edge (LE) Dimming

Older Dimmer designs used an electronic switching device called a Triac. A triac can be turned ON at any point in the mains cycle, but once turned ON, they cannot be turned OFF without reducing the load current to zero. Due to this limitation, triac dimmers can only perform *Leading Edge* (LE) phase control, where the triac turns the power to the load on part way through the cycle and it then remains ON until the current drops to zero as the voltage drops to zero.

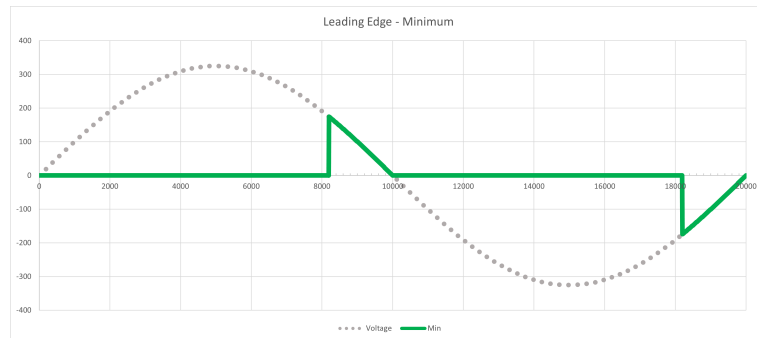
To set a high brightness, the triac is turned on right at the start of the cycle, shortly after the zero crossing, and then allows power through the load for the whole cycle.

The diagram below shows the resulting waveform for a Leading Edge dimmer set to maximum where the maximum conduction is 8 ms. In the positive half-cycle of the mains, the triac remains off until turning on at 2 ms after zero crossing. It then remains on for the 8 ms remaining until the zero crossing at 10 ms. The triac remains off until 2 ms after the zero crossing, when it turns on and remains on for 8 ms until the next zero crossing.



To reduce the brightness, the triac is turned on later in the cycle, some time after the zero crossing. The longer the delay, the lower the conduction time, and the lower the brightness.

The diagram below shows the resulting waveform for a Leading Edge dimmer set to minimum, where the minimum conduction is 1.8 ms. In the positive half-cycle of the mains, the triac stays off until 8.2 ms after zero crossing and turn on for the remaining 1.8 ms until zero crossing at 10 ms. It repeats the same in the negative half-cycle.



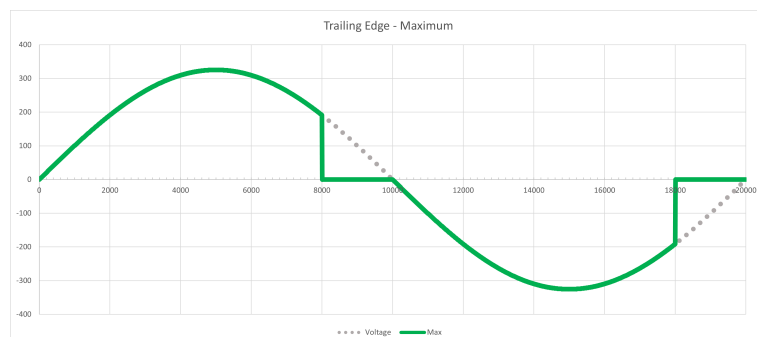
This means depending on where in the cycle the triac turns on, there can be a very large step change in the voltage applied to the load. At 50 % conduction, the step change is equal to the peak of the mains (340 V in a 240 V rms system).

This works well for older style loads such as incandescent lamps and for inductive loads like iron-core low voltage lighting transformers or motors, which naturally limit the inrush current. It is not good for electronic loads such as electronic transformers or LED lighting, which have a capacitance at the input, as this capacitor looks like a short circuit to the triac, causing large inrush current spikes.

## Trailing Edge (TE) Dimming

To overcome this problem, instead of a triac, modern dimmers such as the C-Bus Digital Dimmers use a type of electronic switch called a MOSFET. MOSFETs can be turned both ON and OFF at any point in the mains cycle.

The ability to turn OFF at any point in the mains cycle without waiting for the current to reduce to zero allows MOSFET dimmers to operate in *Trailing Edge* (TE) phase control, where the power to the load is turned ON at zero crossing, and turned OFF after the desired conduction time.



Trailing Edge phase control is ideal for electronic loads like LED lighting, since turning power to the load ON at zero voltage means the inrush current is minimized.

Trailing Edge dimmer is not recommended for inductive loads, since an inductance resists rapid changes in current, resulting in generation of high voltage spikes when the load is switched off.

As a result, Trailing Edge dimmers have a protection mechanism whereby they will shut down if voltage spikes are detected, usually due to an incompatible inductive load being connected.

## Universal Dimmers

Since MOSFETs can turn ON and OFF, MOSFET dimmers can also be designed to operate in Leading Edge mode.

A dimmer which can operate in both modes is often referred to as a *universal* dimmer, which usually start in trailing edge mode (since this is the correct mode for most modern load types), and then switch automatically to leading edge mode if over voltage spikes are detected. Automatic mode selection is not always reliable and often results in erratic or inconsistent behavior.

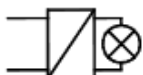
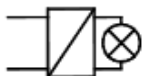
## C-Bus Digital Dimmers

The C-Bus Digital Dimmers are Universal, in that they can operate in either TE or LE mode. However, to avoid the problems with automatic mode selection in traditional Universal dimmers, the correct mode selection is made by the installer, and once set, will not automatically change. The mode for each channel is displayed on, and can be selected from the front panel button for the channel, as well as from the SpaceLogic C-Bus Commission software.

If the mode selected on any channel is unsuitable for the connected load, the channel will simply shut down, and indicate an error via the channel's front panel indicator.

## Compatible Loads

The C-Bus Digital Dimmers are compatible with a wide range of dimmable low-voltage lighting, as long as they operate within the specified channel ratings and built-in protection mechanisms.

| Load Type                                                                                                                                           | Symbol                                                                                         | Required Mode | 5508D1D Rating                                                                                                     | 5504D2D Rating                                                                                                     | 5504DHD Rating                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Dimmable LED Lamps and luminaries                                                                                                                   | LED                                                                                            | TE            | 1 A<br>230 W (230 V)<br>110 W (110 V)                                                                              | 2 A<br>460 W (230 V)<br>220 W (110 V)                                                                              | 5 A*<br>1150 W (230 V)<br>550 W (110 V)                                                                            |
| Incandescent/Halogen Lamps                                                                                                                          |               | TE            | 1 A<br>230 W (230 V)<br>110 W (110 V)                                                                              | 2 A<br>460 W (230 V)<br>220 W (110 V)                                                                              | 5 A*<br>1150 W (230 V)<br>550 W (110 V)                                                                            |
| Dimmable Electronic Transformers with ELV Halogen Lamps, LED Lamps, or LED Strip**                                                                  |               | TE            | 1 A<br>230 W (230 V)<br>110 W (110 V)                                                                              | 2 A<br>460 W (230 V)<br>220 W (110 V)                                                                              | 5 A*<br>1150 W (230 V)<br>550 W (110 V)                                                                            |
| Dimmable Iron-Core Transformers with ELV Halogen or LED Lamps                                                                                       |               | LE            | 1 A<br>230 W (230 V)<br>110 W (110 V)                                                                              | 2 A<br>460 W (230 V)<br>220 W (110 V)                                                                              | 5 A*<br>1150 W (230 V)<br>550 W (110 V)                                                                            |
| Dimmable LED Lamps and luminaries<br><br>Dimmable Electronic Transformers with ELV Halogen Lamps, LED Lamps, or LED Strip<br><br>(Special Cases***) | LED<br><br> | LE***         | 43 W<br>(230 V/PF=0.95)<br><br>20 W<br>(230 V/PF=0.8)<br><br>22 W<br>(110 V/PF=0.95)<br><br>10 W<br>(110 V/PF=0.8) | 86 W<br>(230 V/PF=0.95)<br><br>40 W<br>(230 V/PF=0.8)<br><br>43 W<br>(110 V/PF=0.95)<br><br>20 W<br>(110 V/PF=0.8) | 43 W<br>(230 V/PF=0.95)<br><br>20 W<br>(230 V/PF=0.8)<br><br>43 W<br>(110 V/PF=0.95)<br><br>20 W<br>(110 V/PF=0.8) |

\*Total load across any 2 adjacent channels must be limited to 6 A / 1380 W (230 V) / 660 W (110 V). For more details, refer [Flexible Loading](#), page 17.

\*\* LED strip drivers are limited by inrush current. For more details, refer to the section [Overcurrent Protection](#), page 19.

\*\*\* For more information, refer [Using LE Mode with Electronic Loads](#), page 14.

**IMPORTANT:** It is recommended that motor loads such as exhaust fans or ceiling sweep fans be controlled by an appropriate C-Bus Relay rather than a dimmer channel.

### NOTICE

#### RISK OF EQUIPMENT DAMAGE

- If replacing an older C-Bus dimmer such as an L5508D1A, L5504D2A, L5504D2U, or L5104D5 with a new C-Bus Digital Dimmer, it is important to check that the loads already connected are compatible with the new dimmers, both in relation to channel rating and load type. In some cases, existing loads may not be suitable, or may need to be split across multiple channels or multiple dimmers.
- Never disable the Load Detection Mode before energising a new dimmer channel connected to an existing installation where the load type is unknown.

**Failure to follow these instructions can result in equipment damage.**

## Using LE Mode with Electronic Loads

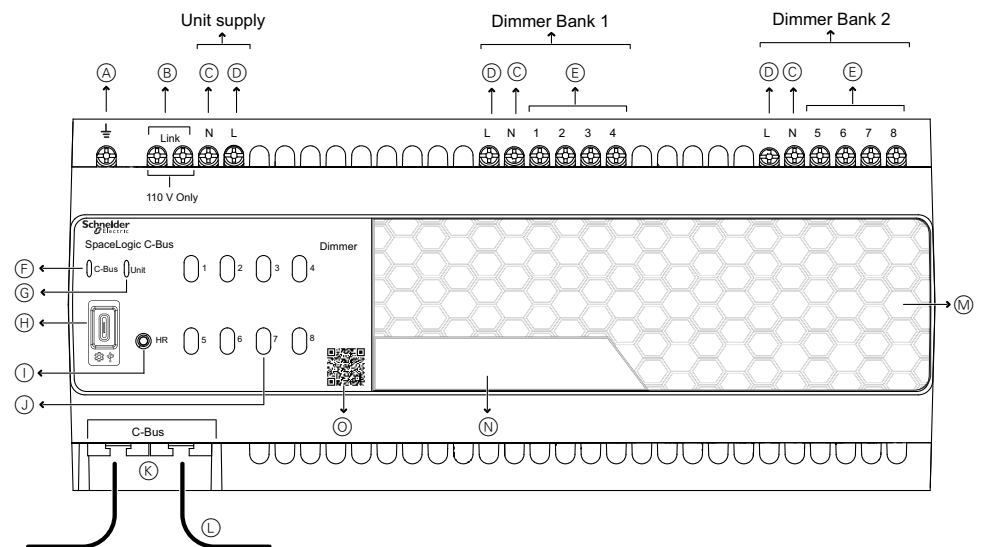
To utilise the full rating of the dimmer, Electronic/Capacitive load types such as LEDs must run in TE mode.

In some situations, such as where a ripple filter may be required upstream from the dimmer, it may be necessary to set the dimmer channels to LE mode, even when using a capacitive load such as an LED.

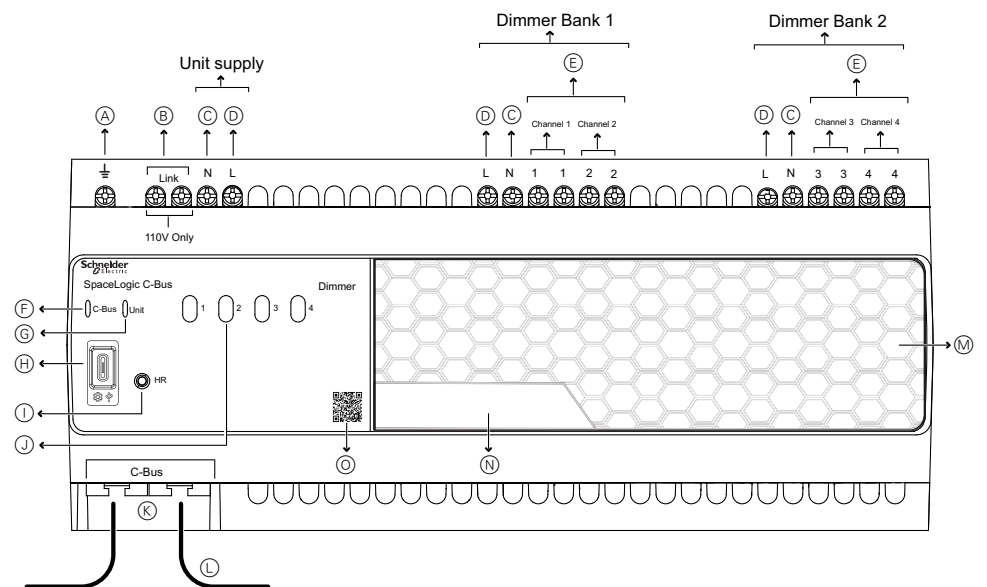
In such cases, the capacitive load creates a high inrush current at turn on every mains half-cycle, which significantly increases the power dissipation of the dimmer. In this configuration, the load rating is significantly reduced as per the table. The rating in this situation is also affected by the power factor of the load (which is an indication of the amount of capacitance it presents to the dimmer channel). Higher power factor is better, and load types with power factor lower than 0.8 should be avoided.

## Hardware Description

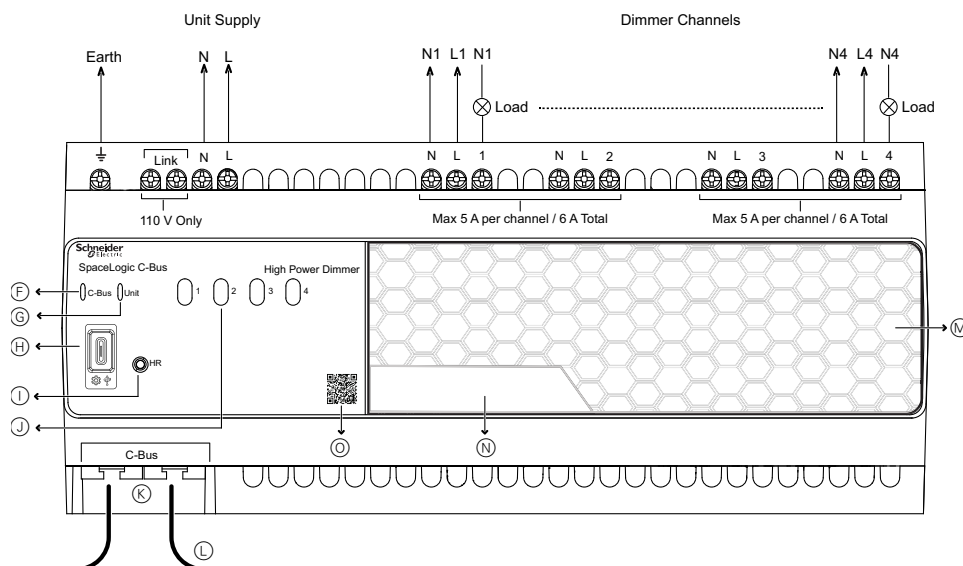
### 8 Channel, 1 A per channel (5508D1D)



### 4 Channel, 2 A per channel (5504D2D)



## 4 Channel High Power (5504DHD)



The below table describes the overview of the elements on the 8 channel, 4 channel and high power dimmer.

|          |                                               |
|----------|-----------------------------------------------|
| <b>A</b> | Earth                                         |
| <b>B</b> | Wire Link Terminal for 110 V                  |
| <b>C</b> | Neutral                                       |
| <b>D</b> | Active Line                                   |
| <b>E</b> | Load Connections                              |
| <b>F</b> | C-Bus Indicator                               |
| <b>G</b> | Unit Indicator                                |
| <b>H</b> | USB (Type-C) Connector                        |
| <b>I</b> | Hard Reset Button                             |
| <b>J</b> | Channel Button Indicators                     |
| <b>K</b> | C-Bus Connectors 2x RJ-45                     |
| <b>L</b> | C-Bus Network Cable                           |
| <b>M</b> | Transparent Removable Cover                   |
| <b>N</b> | Space for Labelling                           |
| <b>O</b> | QR Code for Product Details and Serial Number |



# Project Design and Installation

## Installation Requirements

- Airflow around the unit and the ventilation holes must not be restricted.
- Adequate ventilation should be provided in the enclosure to ensure that the temperature in the enclosure is maintained within the rated operating temperature range of the dimmer.
- If the anticipated ambient temperature levels are higher than operating temperature, it is advised to refer the derating table guidelines to adjust the dimmer load accordingly. For more details, refer [Operating Temperature and Derating](#), page 17.
- The operating calorific values and the heat load must be considered when designing the distribution board.
- Connect only compatible load types to the dimmer.
- The use of external surge protection such as the Clipsal Max9 MX9SP140 or Schneider Electric Acti9 A9L16182 is recommended.

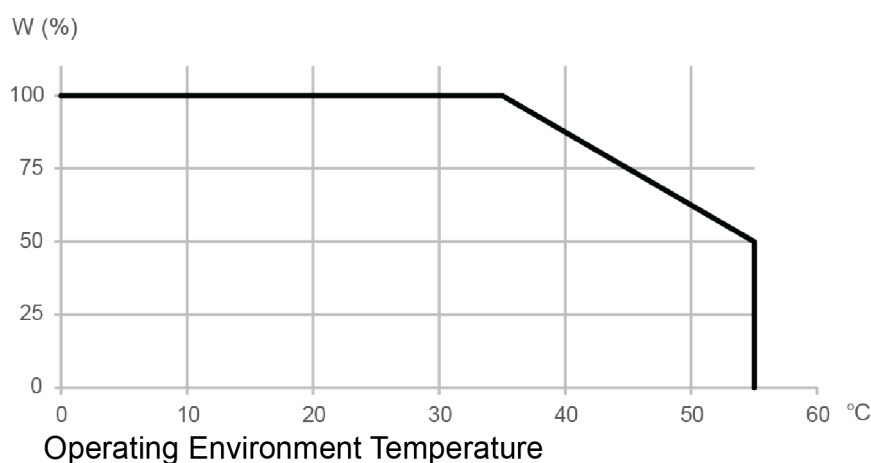
## Operating Temperature and Derating

The dimmer units must be installed in a location where the operating environment temperature (the air temperature immediately surrounding the dimmer unit) is maintained at or below the rated operating temperature.

The dimmer channel ratings are chosen such that during operation with rated load on all channels, and with the unit installed in an environment maintained at the rated operating temperature, the internal components are maintained at temperatures that help to ensure long-term reliability. As a result, there is a trade-off between rated load and rated operating temperature.

The rated load is at 100 % with the temperature of the operating environment equal or below 35 °C, and is reduced linearly according to the chart, to 50 % of nominal at the maximum operating environment temperature of 55 °C.

If the operating environment temperature is greater than 35 °C, the maximum load per channel must be limited in accordance with the chart below. The operating environment temperature must always be maintained below 55 °C.



## Flexible Loading (5504DHD High Power Dimmer only)

Each channel on this model is capable of being loaded up to a maximum of 5 A, as long as the total loading across any 2 adjacent channels does not exceed 6 A.

For example, if the load on Channel 1 is 4.5 A, then the load on Channel 2 must be limited to 1.5 A or less. If Channel 2 is loaded to 1.5 A, then Channel 3 must be limited to 4.5 A (as it is also adjacent to Channel 2).

## Power Loss/Heat Load

The dimmer units produce heat according to the following table. The amount of heat produced by the dimmer unit depends on various factors such as whether the C-Bus Power Supply is enabled, the load on the C-Bus Power Supply, and the load current flowing in each channel. The total heat output of the dimmer unit and all other devices contained within the distribution board must be considered when designing the distribution board, to ensure all devices within the distribution board are maintained within their rated operating temperature range.

|                                                             |      | Power Loss/Heat Load |       |             |
|-------------------------------------------------------------|------|----------------------|-------|-------------|
|                                                             |      | W                    | BTU/h |             |
| Baseline Power (All loads off, C-Bus Power Supply Disabled) |      | 1.3                  | 4.436 | per unit    |
| C-Bus Power Supply Enabled                                  | Adds | 1.0                  | 3.412 | per unit    |
| C-Bus Power Supply fully loaded (200 mA)                    | Adds | 2.9                  | 9.895 | per unit    |
| Dimmer Bank, All channels Off (5508D1D)                     | Adds | 2.0                  | 6.824 | per bank    |
| Dimmer Bank, All channels Off (5504D2D)                     | Adds | 1.0                  | 3.412 | per bank    |
| Dimmer Channel Off (5504DHD)                                | Adds | 0.25                 | 0.853 | per channel |
| 1 A Dimmer Channel @ 1 A load current                       | Adds | 2.25                 | 7.68  | per channel |
| 2 A Dimmer Channel @ 1 A load current                       | Adds | 2.25                 | 7.68  | per channel |
| 2 A Dimmer Channel @ 2 A load current                       | Adds | 4.5                  | 15.35 | per channel |
| HP Dimmer Channel @ 1 A load current                        | Adds | 0.46                 | 1.57  | per channel |
| HP Dimmer Channel @ 2 A load current                        | Adds | 2.10                 | 7.16  | per channel |
| HP Dimmer Channel @ 3 A load current                        | Adds | 3.75                 | 12.79 | per channel |
| HP Dimmer Channel @ 5 A load current                        | Adds | 7.04                 | 24.01 | per channel |

## Protection Features

The C-Bus Digital Dimmers incorporate multiple protection features to ensure safe, reliable operation. The conditions monitored by these features are made available for monitoring via C-Bus Measurement Application and/or the C-Bus Error Reporting Application, as well as via the front panel indicators.

## Thermal Protection

### Over-Temperature Wind-Back

If the load on the dimmer is too high for the environmental and loading conditions, and a dimmer channel temperature rises above 80 °C, the dimmer channel will automatically reduce its brightness to lower the operating temperature below 80 °C. Whilst the brightness is reduced, the condition will be indicated as an *Over Temperature Wind-Back* error condition, both via the front panel indicator for the affected channel (For details refer [Channel Indicator](#), page 33) and via the C-Bus Error Reporting Application.

Once the *Over Temperature Wind-Back* condition is active, if the dimmer channel temperature returns below 80 °C, the operation will return to normal and the *Over Temperature Wind-Back* error condition will be cleared.

Note that where dimmer channels share a common heatsink, an overload on one channel may affect one or more adjacent channels.

## Over-Temperature Shutdown

In the *Over Temperature Wind-Back* condition, if the dimmer channel temperature continues to rise and reaches 100 °C, the dimmer channel will shut down to protect the unit, and the *Over Temperature Shutdown* error condition will be indicated.

Once the over temperature shutdown error condition is triggered, the channel has to be turned off and on again from C-Bus or the front panel, to clear the error and try to restart.

## Thermal Fuses

The Unit Power Supply, as well as each Dimming Bank (for 5508D1D and 5504D2D) or Dimmer Channel (5504DHD only) has a thermal fuse which is required for safety certification. These fuses have a trip temperature well in excess of any temperature that would occur in normal circumstances and will only trip in the event of a serious malfunction or other external source of heat. If these fuses trip, the unit will need to be replaced.

## Overcurrent Protection

Each dimmer channel monitors for overcurrent conditions on a cycle by cycle basis. If the current flowing in the load is too high, the channel will switch off within a fraction of a millisecond, to protect the dimmer channel.

For the 5508D1D and 5504D2D, this limit is approximately 10 A, whereas for the 5504DHD the limit is approximately 40 A. This means that the 5504DHD High Power Dimmer is more tolerant of loads with high inrush such as LED Strip Drivers.

During startup, short term overcurrent conditions can be normal, (example, the cold-filament inrush with an incandescent lamp, or the initial inrush of a dimmable LED strip driver). The dimmer firmware allows for this and automatically performs a certain number of restarts, however if the overcurrent condition persists for too long, the channel will shut down and indicate an *Overcurrent* condition, both via the front panel indicator for the affected channel (For details refer [Channel Indicator](#), page 33) and via the C-Bus Error Reporting Application. If an overcurrent condition is triggered, the channel has to be turned off and on again from C-Bus or the front panel, to clear the error and try to restart.

## Incompatible Load

If the dimmer channel identifies that the connected load is incompatible with the selected mode or load profile, the channel will turn off and indicate *Incompatible Load Error* via the channel indicator. This can often be corrected by configuring a custom load profile with a lower maximum conduction time (for details refer [Load Profiles](#), page 34).

## Mode Error

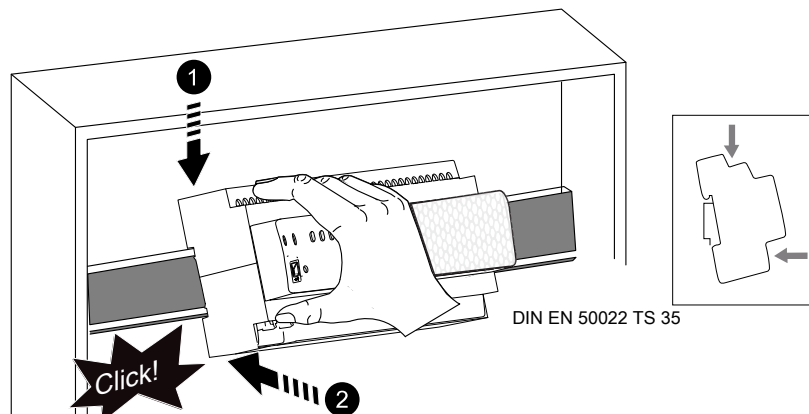
In TE mode, if inductive spikes are observed while dimming (suggesting a presence of an inductive load), the channel will shut down and indicate mode error. If this occurs, channels with inductive loads must be changed to LE mode. If this error condition occurs on channels with other load types, it may be due to excessive inductance in the external supply connection, such as an isolation transformer, or ripple filter.

**NOTE:** When operating a channel in LE mode with an electronic (capacitive) load, the losses are significantly increased, meaning that the rating is significantly lower. Refer to the rating table in [Compatible Loads](#), page 13.

# Mounting and Wiring

## Mounting

The dimmer should only be installed horizontally on the mounting rails (type DIN EN 50022 TS 35) inside a distribution board.



## Electrical Connection

### ⚠️⚠️ DANGER

#### HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

- This product must be installed and serviced by appropriately qualified and/or licensed professional in accordance with the local wiring rules
- Isolate the electrical supply before doing any work on the product
- Ensure that the product has been correctly installed and tested for safe operation before reconnecting the electrical supply
- Do not use this product for any other purpose than specified in this instruction
- Pay attention to the specifications and wiring diagrams related to the installation

**Failure to follow these instructions will result in death or serious injury.**

The dimmer units are capable of handling up to :

- 5508D1D : 8 A across 8 dimmed channels (1 A per channel)
- 5504D2D : 8 A across 4 dimmed channels (2 A per channel)
- 5504DHD : 12 A across 4 channels (maximum of 5 A on any channel, maximum 6 A total on any 2 adjacent channels)

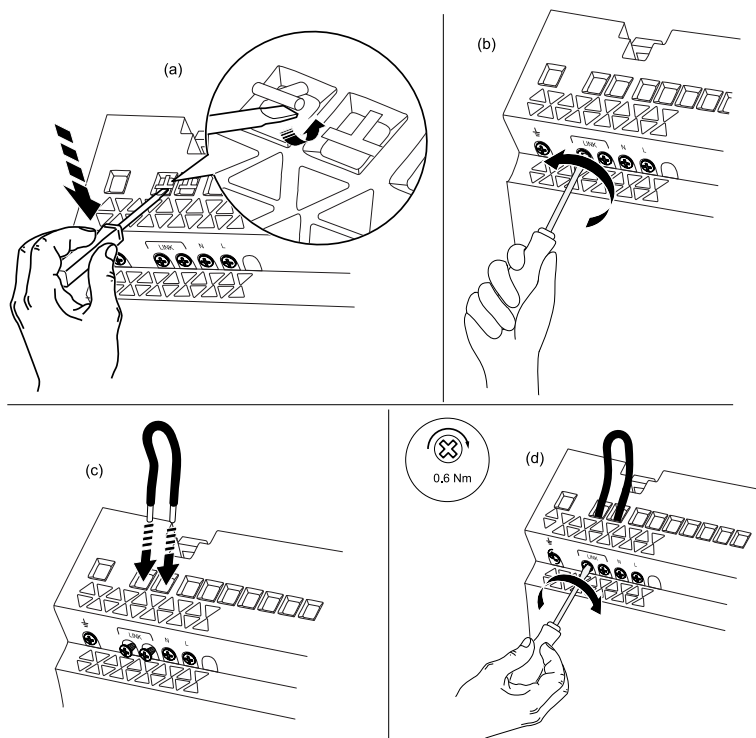
When installing the units:

- The dimmer unit requires mains power provided via the L/N/E connections at the top left, as well as power to the dimming banks (5508D1D /5504D2D) or dimmer channels (5504DHD) in order to operate the dimmer channels.
- All terminals on the dimmer unit are suitable for cable up to 4 mm<sup>2</sup> when using bare cable or 2 x 2.5 mm<sup>2</sup> if using ferrules.
- Make sure while terminating cables into the terminals, that no loose strands of copper fall into unit or enter any aperture other than the terminal.
- The terminals require a PH1 Phillips driver or a 0.8 x 4 flat blade screwdriver.
- Torque must not exceed 0.6 Nm when tightening terminal screws.

- Rubber bungs are supplied for unused RJ45 connectors, to stop foreign bodies from entering the unit. Always install these bungs when the unit is mounted inside a mains rated enclosure.

### 110 V Electrical Wiring

To use the dimmer with a 110 V unit supply, break the T-Block terminal using a screw driver and connect a wire link between terminals.



## ⚠ WARNING

### EQUIPMENT DAMAGE HAZARD

Do not break T-Block terminal and/or insert a wire link unless using the dimmer with 110 V unit supply voltage.

**Failure to follow these instructions can result in death, serious injury, or equipment damage.**

## NOTICE

### EQUIPMENT DAMAGE HAZARD

When breaking the T-Block in the terminal, be careful to not damage other terminal blocks.

**Failure to follow these instructions can result in equipment damage.**

## Phase Independence

The 5508D1D and 5504D2D have two separate Dimmer Banks which each require separate *Active* and *Neutral* connections.

On these variants, each Dimmer Bank may be connected to a different phase.

The 5504DHD High Power Dimmer has independent *Active* and *Neutral* connections per channel, meaning each channel may be connected to any phase.

On all models, the Unit Power Supply may be connected to any phase.

For details refer, 8 channel dimmer.

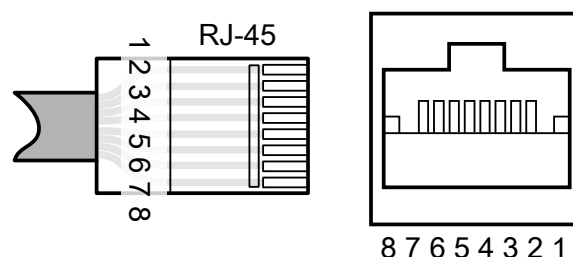
## C-Bus Network Connection

Connection to the C-Bus network is made via one or both of the RJ45 connectors. The two RJ45 connectors are internally connected to allow the C-Bus network to be looped through the unit. See below illustration for pinouts and cable conductor assignments.

Use only C-Bus certified cable, which is identifiable by its pink sheath.

For C-Bus cable, refer:

- Catalogue number: 5005C305B (solid)
- Catalogue number: 5005C305BST (stranded)

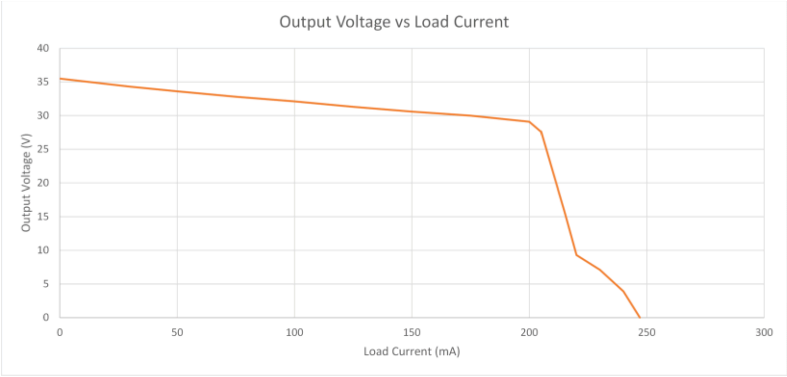


| Pin | Wire Colour<br>Compliance to TIA/EIA 568A wiring standard | Signal             |
|-----|-----------------------------------------------------------|--------------------|
| 1   | Green & White                                             | Remote ON          |
| 2   | Green                                                     | Remote ON          |
| 3   | Orange & White                                            | C-Bus Negative (-) |
| 4   | Blue                                                      | C-Bus Positive (+) |
| 5   | Blue & White                                              | C-Bus Negative (-) |
| 6   | Orange                                                    | C-Bus Positive (+) |
| 7   | Brown & White                                             | Remote OFF         |
| 8   | Brown                                                     | Remote OFF         |

## C-Bus Power Requirements

Adequate C-Bus Power Supply units must be installed or enabled to support the total consumption of all C-Bus devices connected to the C-Bus Network segment.

C-Bus Power Supplies are specially designed to share load when distributed around the C-Bus Network. To achieve this, the integrated C-Bus Power Supplies have an output impedance of approximately 32  $\Omega$  up to their maximum current of 200 mA. Once the maximum current is exceeded, the output impedance increases sharply to limit the current to a safe level in the event of a fault in the C-Bus Network wiring.



There must not be more than 2000 mA of total C-Bus Power Supply capacity enabled on a single C-Bus Network segment as this is the limit of the current carrying capacity of the C-Bus Network cable.

The dimmer unit includes a switchable inbuilt C-Bus Power Supply, which can supply up to 200 mA to the C-Bus Network when enabled. By default, the inbuilt Power Supply is disabled, but can be enabled (or disabled) via the front panel.

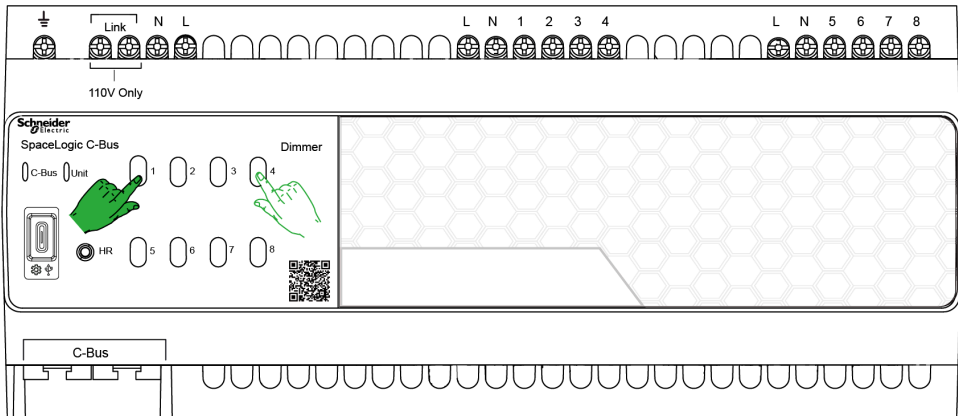
The SpaceLogic C-Bus Commission software provides you with information whether the inbuilt Power Supply is enabled or disabled in each unit, as well as a summary of the total network power capacity and consumption. This can be helpful in determining the Power Supply requirements of a particular network.

**NOTE:** The dimmers draw 20 mA from the C-Bus Network when main power is not connected or when the C-Bus Power Supply is not enabled.

## Enabling/Disabling the C-Bus Power Supply from the Front Panel

The integrated C-Bus Power Supply can be enabled or disabled via the front panel by performing a short press on the top right channel button whilst holding down the top left channel button to toggle the enabled state. After the desired setting for the C-Bus Power Supply has been made, the function to change the setting from the front panel may be disabled via the SpaceLogic C-Bus Commission software to prevent further changes.

**NOTE:** Changing the Power Supply state will temporarily disrupt C-Bus Network communications, which can take several seconds to recover, depending on the network configuration.



The enabled status of the integrated C-Bus Power Supply is shown by the colour of the Unit Indicator.

| Unit Indicator                                                                      | Function                       |
|-------------------------------------------------------------------------------------|--------------------------------|
|  | C Bus Power Supply is enabled  |
|  | C Bus Power Supply is disabled |



Refer Unit Indicator, page 32 for more details.

The state of the network voltage is shown by the **state** of the C-Bus Indicator.

- If the C-Bus Indicator is flashing or off, the network voltage is too low, meaning there is insufficient C-Bus power.
  - In this case, additional power supplies may need to be enabled.
- If the C-Bus Indicator is steady, the network voltage is sufficient.

The state of the C-Bus system clock is shown by the colour of the C-Bus Indicator.

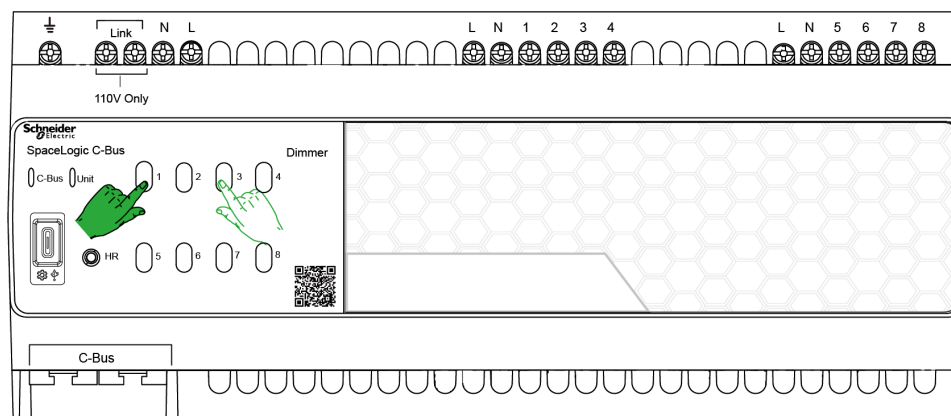
| C Bus Indicator                                                                                                                                                                                   | Function                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <br>or<br><br>(Green or Yellow) | C Bus clock signal has been detected     |
|                                                                                                                  | No C- Bus clock signal has been detected |

Refer C-Bus Indicator, page 33 for more details.

## Enabling/Disabling C-Bus System Clock Generator from the Front Panel

A C-Bus clock signal is required to synchronize communication on the C-Bus Network. The dimmer unit (and most other C-Bus DIN Rail mounted devices) can be configured to generate the C-Bus system clock. It is recommended that the C-Bus system clock generator function be enabled in three units on each network segment.

If a C-Bus system clock is required on the dimmer unit, it can be enabled or disabled from the front panel by performing a short press on the top second-to-right channel button whilst holding down the top left channel button. This toggles the enabled state of the C-Bus system clock generator in the unit.



After the desired setting for the C-Bus system clock generator, the function to change the setting from the front panel may be disabled via the SpaceLogic C-Bus Commission software to prevent further changes.

# Using the Device

## First Power Up/Load Detection Mode (TE/LE)

To ensure consistent operation throughout the life of the dimmer, the dimmer channels do not have automatic mode selection.

By default, all dimmer channels are set to **Load Detection Mode**. In this mode, the dimmer channel operates in Trailing Edge mode with a very low minimum level, and a very slow soft turn on. This is to ensure that when turned on for the first time, the dimmer can detect iron core loads safely and reliably, avoiding possible damage to the dimmer.

**For normal operation, the correct dimming mode for each channel needs to be confirmed.** Confirming the dimming mode turns off Load Detection mode and selects the normal Trailing Edge or Leading Edge mode appropriately.

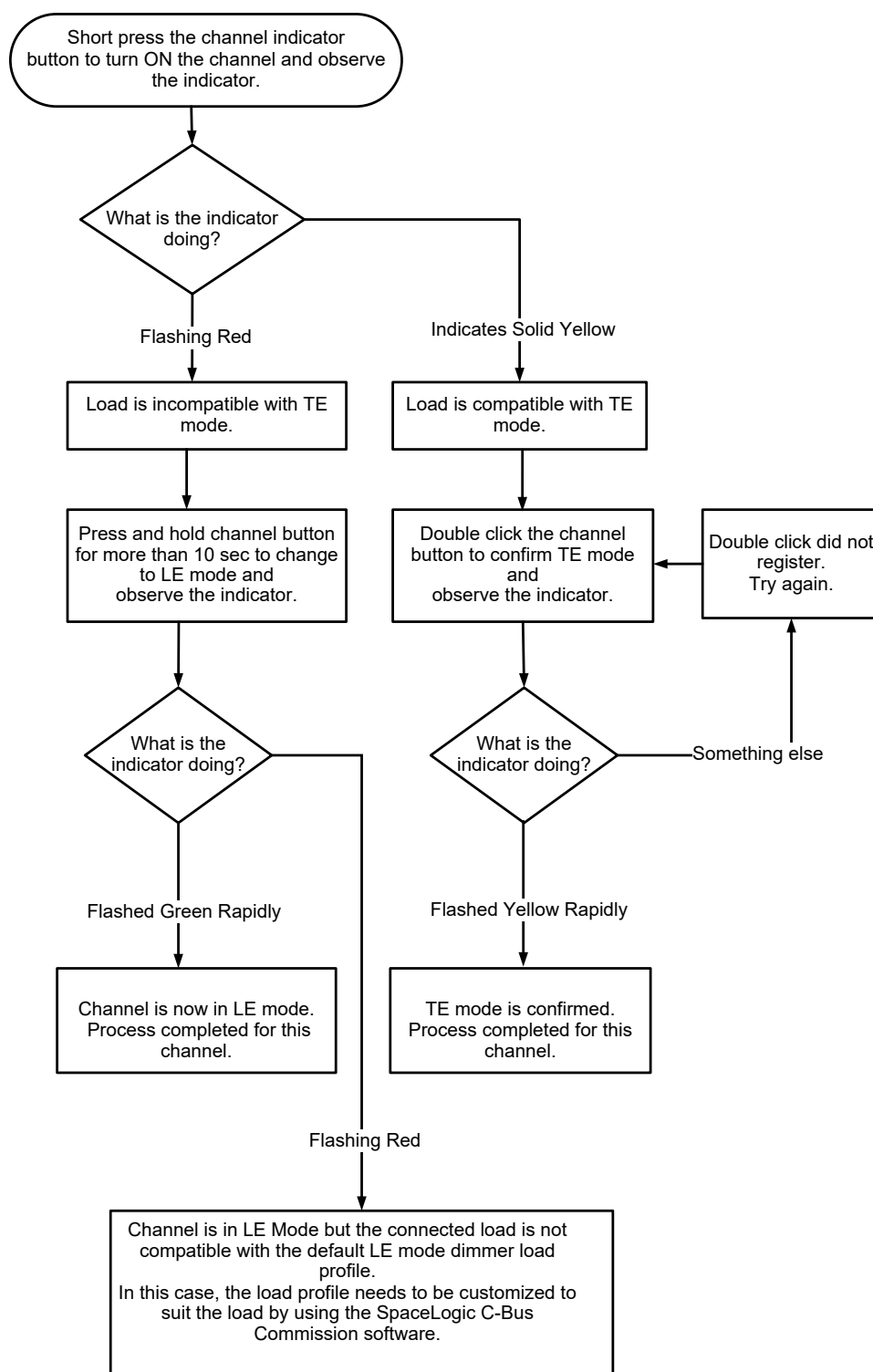
For more details refer, selecting the right dimming mode, page 27 and load profiles, page 34.

## Confirming the Dimming Mode

The purpose of the Load Detection Mode and the Mode Confirmation process is to ensure compatibility of the selected dimming mode with the intended load type. If the mode confirmation is performed without the load attached or with a different load, subsequent operation in TE mode with some types of iron-core load types may cause damage to the dimmer channel.

As part of the initial installation, the correct mode for each channel must be confirmed.

Refer the below flow chart:



**NOTE:** If it is necessary to change from LE mode back to TE mode, hold the channel button for more than 10 seconds. Once done, Load Detection Mode is re-enabled, and the Mode Confirmation process must be repeated.

## Power-Up Load Status

While designing an installation, it is important to understand the power-up behaviour of the system.

The following points should be considered:

- The SpaceLogic C-Bus Commission software allows the dimmer channels to be configured to restore any fixed level, or restore to their previous level upon restoration of power.
  - By default, all channels are set to restore to their previous state, with the stored previous state being OFF.
- On restoration of power, the dimmer waits approximately 5 seconds for power to stabilize before restoring the channel states.
- If a dimmer channel loses power, upon recovery of power, the channel is held OFF for the *Power On Delay* time programmed for that channel (default set to 0).

## Dimmer Channel Control

The output state of a dimmer channel can be changed by:

- A lighting command from C-Bus.
- Pressing one of the channel indicator buttons (local override).
- Changing the state of the remote override inputs.

## Priority of the Operating Modes

The various sources of control listed above are described in detail below. The table below shows the relative priority of these sources in determining the final state of the dimmer channel:

| Function                       | Priority    | Result                                                                 |
|--------------------------------|-------------|------------------------------------------------------------------------|
| Remote OFF                     | 1 (highest) | Forces Channel Off                                                     |
| Unit Power Stabilisation Delay | 2           | All channels off for first 5 s after unit power up.                    |
| Remote ON                      | 3           | Forces channel On                                                      |
| Local override                 | 4           | Toggles the channel                                                    |
| Channel Power On Delay         | 5           | Holds channel off for configured duration after channel power recovery |
| Logic                          | 6           | Depends on Logic configuration and state of Logic Groups               |
| Warn Before Off (WBO)          | 7           | Holds channel on at WBO level for WBO Duration if WBO active           |
| C-Bus Level                    | 8           | Channel Level from C-Bus                                               |

## Control from C-Bus

In order to be used, a dimmer channel must be assigned a group address from one of the four available Lighting Application addresses that can be configured in the dimmer unit. The same group address can be assigned to multiple channels.

**NOTE:** Unused channels should be left without a group address configured.

More advanced settings can be configured to modify the conditions required for the dimmer channel to turn on. The advanced settings are:

- Logic
- Power-On Delay
- Warn Before Off

## Power on Delay

Each channel has a configurable power-on delay parameter, which can be used to delay the recovery of the channel state upon restoration of power after a power loss. This can be used to help manage inrush for the installation.

## Min/Max Scaling

The Min/Max scaling feature takes the input control level range of 1-255 and scales it to the range Min-Max where Min and Max are proportional to the minimum and maximum conduction values defined by the load profile for the channel.

- An input control level of 0 is always 0 (off).
- An input control level of 1 is scaled to Min.
- An input control level of 255 is scaled to Max.

This allows the desired working range for control from a user's point of view to be managed independently of the minimum and maximum conduction settings in the load profile.

- Minimum/Maximum conduction in the load profile is an adjustment to adjust the dimmer to the lamp's needs.
- Min/Max scaling is an adjustment to meet the end user's needs.

Example: The same downlight with the same load profile may be used in two different rooms. If the lamp flickers at the default maximum conduction, then you should reduce the maximum conduction in the load profile to a value where it no longer flickers. If the lamp is now too bright for the user's liking in one room but not in another, change the Max in the Min/Max scaling for the channel feeding the room where it is too bright. This allows the user to set the channel to 100% and get the (different) desired lower maximum brightness in that room, while retaining the higher maximum brightness in the other room.

## Warn Before Off

Warn Before Off is an optional courtesy feature of the Dimmer. It can be enabled by an Enabled Control Group and the duration can be set up to 15 min.

When Warn Before Off is enabled for a channel, and a command is received from C-Bus to turn the channel OFF, the channel does not immediately turn off. Instead it:

- sets the channel level to the configured Warn Before Off level, to warn any occupants that the light will turn off shortly.

- starts a countdown timer set for the Warn Before Off time:
  - When the timer expires, the channel turns OFF.
  - If another OFF command is received for the channel before the timer expires, the channel cancels the timer and turns off immediately.

## Logic

The Logic feature allows the level of up to four additional *Logic* group addresses to be combined with the dimmer channel's primary group address to determine the final level that controls the dimmer channel's output level. There are four logic groups available in the dimmer unit, and each channel can be combined with any combination of these.

In dimmers, there are two logic functions available :

1. In MIN logic, the output level takes the minimum value of the primary lighting group and the states of all logic groups which that channel is configured to be combined with.
  - a. This is useful as a way of limiting the brightness at different times of day. The primary group can be used to control the level and state, and the logic group can be used to limit the brightness. For example, a schedule can set the logic group to a lower level after dark so that lights turned on during the night only come on at a low level. The schedule would then also set return the logic group level to maximum at sunrise so that the lights starts to function normally again.
2. In MAX logic, the output level takes the maximum value of the primary lighting group and the states of all logic groups which that channel is configured to be combined.

## Local Override

The channel indicator buttons on the front of the unit, toggle the corresponding channel on and off, providing a local override capability. This can be disabled in software. Each button illuminates when its respective channel is in the ON state. The channel indicator buttons perform different functions depending on how they are pressed.

A long-press on any channel indicator button for one second or more, returns all channels to the C-Bus Network level.

**NOTE:** Double short-press and long-press operations only apply when the unit/channel is in local override mode.

By default, any C-Bus commands received by the unit will override local toggle changes. In this case, only the channel associated with the received commands will revert to the C-Bus Network state. This option can be disabled in the software settings.

## Remote Override

The extra two pairs of the C-Bus Network cable provide high priority override functions for most C-Bus Output units.

- The Green & Green/White pair is used for the Remote ON function.
- The Brown & Brown/White pair is used for the Remote OFF function.

Connecting either pair to the C-Bus negative pair activates the corresponding function.

Each dimmer channel can be individually included or excluded from either or both of the Remote ON and Remote OFF functions. By default, all channels are included in both. This setting can be changed using the SpaceLogic C-Bus Commission software.

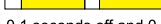
The SpaceLogic C-Bus Network Automation and Application Controllers provide facilities to control these overrides.

**IMPORTANT:** During the installation, it is important to make sure these pairs are not left unterminated, as this may cause the override functions to be triggered by accident.

# Indicators

## Unit Indicator

The unit indicator shows status related to the unit itself, such as the internal temperature, configuration issues, firmware update etc.

| Action                     | LED Indication                                                                                                                | Status                                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Green LED is ON            |                                              | <ul style="list-style-type: none"> <li>Unit powered.</li> <li>C-Bus Power Supply enabled.</li> </ul>                                                                   |
| Yellow LED is ON           |                                              | <ul style="list-style-type: none"> <li>Unit powered.</li> <li>C-Bus Power Supply disabled.</li> </ul>                                                                  |
| LED OFF                    |                                              | No power to device.                                                                                                                                                    |
| Red LED is ON              |                                              | <ul style="list-style-type: none"> <li>Unit powered.</li> <li>Configuration error.</li> </ul>                                                                          |
| Slow Flash Red, Green      | <br>0.5 seconds red and 0.5 seconds green    | <ul style="list-style-type: none"> <li>Internal temperature has exceeded over-temperature threshold.</li> <li>C-Bus Power Supply enabled.</li> </ul>                   |
| Slow Flash Red, Yellow     | <br>0.5 seconds red and 0.5 seconds yellow   | <ul style="list-style-type: none"> <li>Internal temperature has exceeded over-temperature threshold.</li> <li>C-Bus Power Supply disabled.</li> </ul>                  |
| Override Flash Red, Green  | <br>0.1 seconds red and 0.9 seconds green    | <ul style="list-style-type: none"> <li>Internal temperature has exceeded over-temperature threshold.</li> <li>Override active, C-Bus Power Supply enabled.</li> </ul>  |
| Override Flash Red, Yellow | <br>0.1 seconds red and 0.9 seconds yellow | <ul style="list-style-type: none"> <li>Internal temperature has exceeded over-temperature threshold.</li> <li>Override Active, C-Bus Power Supply disabled.</li> </ul> |
| Override Flash Off, Green  | <br>0.1 seconds off and 0.9 seconds green  | <ul style="list-style-type: none"> <li>Override active.</li> <li>C-Bus Power Supply enabled.</li> </ul>                                                                |
| Override Flash Off, Yellow | <br>0.1 seconds off and 0.9 seconds yellow | <ul style="list-style-type: none"> <li>Override active.</li> <li>C-Bus Power Supply disabled.</li> </ul>                                                               |
| Fast Flash, Green          | <br>0.2 seconds green and 0.2 seconds off  | Secondary firmware update in progress.                                                                                                                                 |



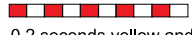
## C-Bus Indicator

The C-Bus Indicator shows the status of the C-Bus Network voltage and communications. The colour of the indicator relates to the C-Bus Network communications status, whereas the state of the indicator (solid, flashing etc) relates the C-Bus Network voltage.

| Action              | LED Indication                                                                    | Status                                        |
|---------------------|-----------------------------------------------------------------------------------|-----------------------------------------------|
| Solid(not flashing) | Regardless of color                                                               | C-Bus voltage OK.                             |
| Steady Flash        |                                                                                   | C-Bus voltage marginal (15-20 V).             |
| Short Flash         |                                                                                   | C-Bus voltage critical (<15 V).               |
| Red                 |  | No C-Bus clock detected, host online.         |
| Green               |  | C-Bus clock detected, host online.            |
| Yellow              |  | C-Bus clock detected, host offline.           |
| Off                 |  | No C-Bus voltage present or unit not powered. |

## Channel Indicator

The channel indicators show the operating mode of the channel as well as providing feedback on any configuration changes or error conditions.

| Action                                        | LED Indication                                                                                                                | Status                                                                                                      |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Short-press on the channel<br>(< 0.5 seconds) |                                            | Channel On, Leading Edge mode.                                                                              |
|                                               |                                            | Channel On, Trailing Edge mode.                                                                             |
| Slow Flash, Red                               | <br>0.5 seconds red and 0.5 seconds off    | Channel shutdown due to incompatible load or mode error.                                                    |
| Off                                           |                                            | Channel Off.                                                                                                |
| On, Red                                       |                                            | Channel Offline, normally due to no mains supply to channel or dimmer channel firmware requiring an update. |
| Fast Flash, Red                               | <br>0.2 seconds yellow and 0.2 seconds off | Channel shutdown due to over current.                                                                       |
| Fast Flash, Yellow                            | <br>0.2 seconds yellow and 0.2 seconds off | Channel shutdown due to over temperature condition.                                                         |
| Slow Flash, Yellow                            | <br>0.5 seconds yellow and 0.5 seconds off | Channel operating at reduced setting due to over temperature condition.                                     |

# Configuring the Device

Configuration and commissioning of the C-Bus unit is done by the C-Bus connection using the *SpaceLogic C-Bus Commission* software. To download the software [click here](#).

The unit must be powered for configuration, however power may be provided by the USB-C connection for programming via C-Bus (without an energized mains connection).

**NOTE:** Dimmer channels will not be functional when powered solely from the USB-C connection.

## Customising the Load Profile and Dimming Curve

LED load types can vary dramatically in their behaviour. Models/Types can vary in their dimming range, minimum and maximum conduction, and turn on delay. Traditional dimmers have fixed the dimmer operating limits, which makes it difficult to achieve the optimum performance for all of the different load types that might be encountered.

The C-Bus Digital Dimmers feature highly customisable **Load Profiles** and configurable **Dimming Curves**, which can be configured and selected using the SpaceLogic C-Bus Commission software.

### Load Profiles

A Load Profile adjusts certain settings within each dimmer channel which tailor the dimming behavior to improve the end to end dimming experience.

The default inbuilt load profiles for TE and LE mode are conservative settings which are chosen to work well with most *typical* loads, but cannot be modified.

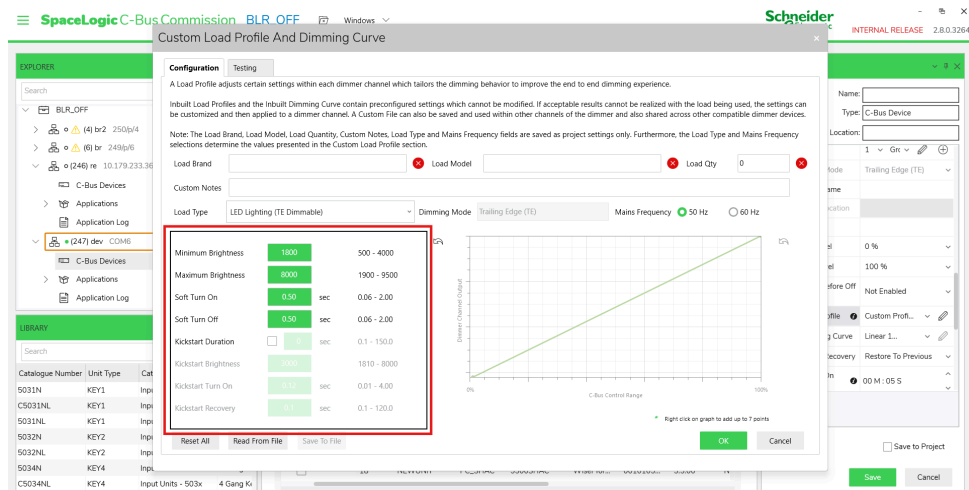
If acceptable results cannot be realized with the inbuilt load profiles, a Custom Load Profile may be defined and applied to a dimmer channel.

Each dimmer supports up to four Custom Load Profiles.

Custom Load Profiles can be named, saved, and shared, allowing commonly used settings to be easily re-used.

The Custom Load Profiles allow the minimum and maximum conduction values to be set, to optimise the dimming range for the load. They also allow the soft turn on and soft turn off times to be adjusted, as well as allowing the configuration of a kickstart, which can aid in startup for problematic load types such as LED strip drivers or fan motors.

The below image displays the **Custom Load Profile** from SpaceLogic C-Bus Commission software.



## Minimum Brightness

Minimum Brightness sets the lowest conduction time for the channel. This is the brightness level at which the channel turns on at the start of its soft turn on, and the brightness level that will be set when the C-Bus Group controlling the channel is set to minimum. This should be adjusted to the lowest value where the lamp is operating in a stable manner, and turns on reliably and in a reasonable amount of time.

**NOTE:** Many lamps will behave slightly differently when they have been running for a period of time and have warmed up, so be sure to test a *cold start* by leaving the channel off for a few minutes when experimenting with lower values for this setting.

**NOTE:** Minimum and maximum brightness settings set by the load profile are independent of the Min/Max scaling settings. The Min/Max scaling settings can be used to modify the control range of C-Bus minimum to C-Bus maximum within the operating range set by the minimum and maximum conduction values in the load profile. For more details, refer [Min/Max Scaling](#), page 29.

## Maximum Brightness

Maximum Brightness sets the highest conduction time for the channel. Many lamps will reach their maximum brightness at a conduction time lower than full conduction. The maximum brightness should be set to the value at which no further visible change in brightness is observed as the conduction time is increased.

## Soft Turn-On

Soft Turn On sets the fade rate used by the dimmer when changing between levels with an *Instant* ramp. The value corresponds to the time taken to fade from minimum to maximum when the channel is toggled from off to on. The range of values that can be set is limited depending on the settings for minimum and maximum brightness.

## Soft Turn-Off

Soft Turn Off sets the fade rate used by the dimmer when changing between any on level and off, with an *Instant* ramp. The value corresponds to the time taken to fade from maximum brightness to off.

## Kickstart

Some types of loads have problems starting at low conduction/brightness settings. In the case of an LED lamp, this can show up as the lamp taking a long time to turn on when turned on at a low setting, or taking a long time to respond and then suddenly turning on at a higher level when ramped up slowly from off. In the case of a motor such as a sweep fan, it might take a long time to start moving or not move at all at low settings.

The Kickstart is a feature which, when enabled, uses a higher brightness setting for a short amount of time when transitioning from off to on, which improves the startup behavior of loads with these kinds of problem.

When kickstart is enabled, each time the channel is turned on, if the level is less than the *Kickstart Brightness*, the channel will fade to the *Kickstart Brightness* setting over the duration set by the *Kickstart Turn on* setting. It will then stay at the Kickstart Brightness setting for the duration set by the *Kickstart Duration*, and then fade to the current level over the duration set by the *Kickstart Recovery* setting.

**Kickstart Duration:** Kickstart Duration sets the duration for maintaining the kickstart brightness level before recovering to the C-Bus level.

**Kickstart Brightness:** Kickstart Brightness sets the dimmer kickstart minimum brightness level before when transition from off to on (at any level).

**Kickstart Turn on:** Kickstart Turn on sets the time taken for the transition from minimum brightness to the kickstart brightness at the beginning of the kickstart. The range of values that can be set is limited depending on the settings for minimum brightness and kickstart brightness.

**Kickstart Recovery:** Kickstart Recovery sets the time taken for the transition from the kickstart brightness level to the currently set level of the load at the end of the kickstart duration period.

## Dimming Curve

A Dimming Curve refers to a translation between C-Bus input level and dimmer channel output (Conduction Time/Brightness).

For each Custom Load Profile, a custom Dimming Curve may also be defined.

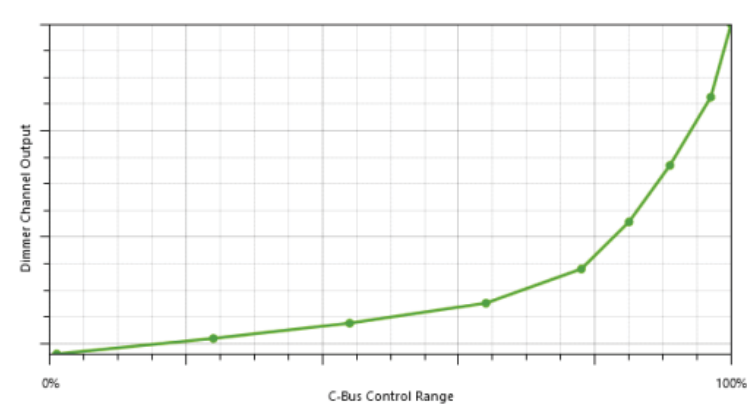
In all dimming curves, an input level of 1 (minimum level on C-Bus) is mapped to minimum brightness and an input level of 255 (maximum level on C-Bus) is mapped to maximum brightness.

With the inbuilt load profiles, a linear dimming curve is used, which means that for example, an input level of 50 % maps to a brightness value 50 % of the way between the minimum brightness and the maximum brightness. This does not always give a good pushbutton dimming experience.

When dimming with a C-Bus pushbutton dimming function, the ramp rate (the change in C-Bus Level vs time) is constant, indicating the change of conduction time is constant during the ramp. However, as the relationship between perceived brightness vs conduction time may not be linear, this may result in a very non-linear change of perceived brightness as you dim.

For example, the brightness may change very fast at the bottom of the dimming range and much more slowly at the top end of the range. In such cases, by defining a custom Dimming Curve which is flatter at the bottom end of the range and steeper at the top, there is better control available at the bottom end where it is needed because as the level ramps up, the change in conduction time is slower at the bottom end, compensating for the faster resulting brightness change, making the resulting rate of change of brightness more constant through the dimming range.

### Example: Custom Dimming Curve



**NOTE:** The Dimming Curve translation is applied after Min/Max scaling. The Min/Max scaling is effectively just limiting the operating range on the horizontal axis of the dimming curve rather than altering the space of the curve by squashing it over a smaller range.

The SpaceLogic C-Bus Commission software includes examples which can be used as starting points for development of Custom Load Profiles for the most common load types :

- LED Lighting (TE Dimmable)
- LED Lighting (LE Dimmable)
- Electronic Transformer (TE Dimmable)
- Electronic Transformer (LE Dimmable)
- Incandescent Lighting (TE Preferred)
- Incandescent Lighting (LE)
- Sweep Fan
- Exhaust Fan

## Testing Custom Load Profile Settings

When connected to a live C-Bus Network, the SpaceLogic C-Bus Commission software also allows changes in the Custom Load Profile settings to be tested in real-time directly from the software, without the need to deploy to a dimmer and switch.

The testing user interface allows the dimmer channel to be turned on and off from the software at different levels, with different settings for minimum and maximum conduction, to rapidly find the best settings for the given load.

The screenshots below shows the process for testing operation:

### Step 1:

Custom Load Profile And Dimming Curve

Configuration Testing

The testing operation is now underway. Step 1 will help you discover and set the minimum and maximum brightness limits to utilise as much of the available dimming range as possible with the load currently connected. Eliminate erratic behaviour of lights at low brightness levels and mask little/no change in response of lights at high C-Bus control levels.

Dimmer: 5504D2D (1) Load Profile: Custom Profile A Dimming Mode: Leading Edge (LE)

Channel: Channel 2 Dimming Curve: Custom Curve A

#### Step 1

This step will help you discover and set the Minimum and Maximum Brightness limits to maximise the dimming range for the lights currently connected to the dimmer channel. Find and set the lowest acceptable brightness level and test the Turn On behavior at that level. Find and set the highest acceptable brightness level and test the Turn Off behavior at that level. To test the dimming response, proceed to Step 2. Otherwise click Save to end the testing operation.

Minimum Brightness  
2200

Maximum Brightness  
7000

Channel:  
Turn On at Minimum  
Turn On at Maximum  
Turn Off

Step 2

Save Cancel

### Step 2:

Custom Load Profile And Dimming Curve

Configuration Testing

The testing operation is now underway. Step 2 will help you test and observe the dimming behaviour within the range defined in Step 1. Confirm the ramp response between the Minimum and Maximum Brightness limits with the load currently connected.

Dimmer: 5504D2D (1) Load Profile: Custom Profile A Dimming Mode: Leading Edge (LE)

Channel: Channel 2 Dimming Curve: Custom Curve A

#### Step 2

From the values set in Step 1, now test the channel dimming behavior using Ramp to Minimum, Ramp to Maximum and Ramp to Off buttons. A C-bus ramp rate value can also be set to further explore the dimming response at different speeds.

Minimum Brightness  
2200

Maximum Brightness  
7000

Ramp Rate: 12 secs  
Channel:  
Ramp to Minimum  
Ramp to Maximum  
Ramp to Off

Step 1

Save Cancel

## Unit Monitoring

The C-Bus Digital Dimmers include extensive monitoring and reporting capabilities, to aid in rapid identification of abnormal operating conditions.

## Measurement Application

The Measurement Application is used for broadcasting values of physical parameters monitored by C-Bus devices.

The measured quantities can be displayed on devices with graphical user interfaces such as eDLTs or Automation Controllers, and can be used as inputs to automation in SpaceLogic C-Bus Automation Controllers.

Measurement Application message contains the following information:

- A device ID to identify the unit which broadcast the message.
- The channel number, which is used to differentiate between multiple measurements that the single device may produce.
- The measured value.
- The units of the measured value (example: Volts, Amperes, Degrees).

The C-Bus Digital Dimmers provide measurements of the following values via the Measurement Application which can be useful for monitoring the health of the unit, or for detecting abnormal conditions:

| Channel Number | Quantity                                    | Units   |
|----------------|---------------------------------------------|---------|
| 0-7            | Channel 1 — Channel 8 Lamp Running Time     | Hours   |
| 128-135        | Channel 1 — Channel 8 Operating Temperature | °C      |
| 252            | Power Supply Load Current                   | Amperes |
| 253            | C-Bus Voltage                               | Volts   |
| 254            | Unit Operating Temperature                  | °C      |

Measurements can be configured to broadcast periodically, by setting the measurement periodic broadcast parameter. This can be set to an interval between 1 and 254 minutes.

**NOTE:** If periodic broadcast is not required, a Trigger Group can be configured to request either individual channels (by using the Action Selector corresponding to the channel requested), or an Action Selector of 255 to request all channels to be updated. This can also be used in conjunction with a regular broadcast.

In addition, individual quantities may be excluded from the regular broadcast by de-selecting them in the Measurement Regular Broadcast options parameter.

The Lamp Running Time measurements can be reset to zero by configuring a Trigger Group and sending a trigger event on that Trigger Group with the action selector set to the channel which is to be cleared, or an Action Selector of 255 to clear all channels simultaneously.

## Error Reporting Application

The C-Bus Error Reporting Application is used to notify and manage error or other status information detected by C-Bus devices.

C-Bus dimmer units can monitor various operational parameters, and can report error or warning conditions using the C-Bus Error Reporting Application.

The reports consist of information on the source, severity and nature of the error or fault condition, and also allow independent tracking and acknowledgement of the most recent and most severe error conditions.

The error reports are monitored by a device such as a SpaceLogic C-Bus Automation Controller, which can:

- Display the status to the user.
- Allow them to acknowledge reports.
- Allow them to clear any latched errors.

A SpaceLogic C-Bus Automation Controller can also make all of this available to a higher layer such as a BMS via BACnet.

There are 5 Error Reporting Modes, to control:

- which error reports are generated
- when they are generated

The mode can either be fixed during commissioning time, or can be changed dynamically by assigning a control group on the Enable Control application. In addition to controlling the generation of error reports using the mode, a Trigger Group may be configured to request an update to ALL error channels at any time.

This can be useful to manage traffic in an installation, where a large number of Error Reporting devices or channels exist.

| Mode | Error Level                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    | Off                                                       | In this mode, neither live reports nor regular reports are generated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1    | All Channels<br>Most Recent Errors only                   | In this mode, live reporting is enabled, meaning a change in error state for any channel will be emitted when it occurs.<br><br>In addition, if the Regular Reporting Interval parameter is configured, the current error state of all channels will be re-sent at the configured interval.<br><br>Setting the Regular Reporting Interval to zero disables the regular reports, but live reporting remains active. This allows timely notification of error status changes but without the traffic overhead of additional regular reports. |
| 2    | All Channels<br>Most Recent Errors and Most Severe Errors | This mode works the same as Mode 1, but also includes tracking of the Most Severe errors for each channel.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3    | Minimum reporting<br>Most Recent Errors only              | This mode works the same as Mode 1, but excludes the individual channel reports.<br><br>This can be useful to reduce the amount of traffic where there are a large number of channels or a large number of devices exist.<br><br>The aggregate channel is included, which is a logical OR of all of the channel error bytes, which still allows channel errors to be monitored.                                                                                                                                                            |
| 4    | Minimum reporting<br>Most Recent and Most Severe Errors   | This mode works the same as Mode 2, but excludes the individual channel reports.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



The following channel error status information is available via the Error Reporting Application:

| Error Channel                                                                                                                                                                                                                 | Error Type          | Error Data Format                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |                                                                                                                               |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------|---|--------------|--------------------------------------------------------------------------------------------------------|---|--------------------|-------------------------------------------------------------------------------------------------------------------------------|---|------------------------|----------------------------------------------------------------------------------------------------------|-----|-----------------|---------------------------------------------------------------------------------------------------------------|---|------------|---------------------------------------------------------------------------------------------------------|---|----------------------------|-------------------------------------------------------------------------------------|---|-----------------------|---------------------------------------------------------------|---|---------------------------|-------------------------------------------------------|
| 0 - 7                                                                                                                                                                                                                         | Channel 1-8 Errors  | <table><tr><th>Bit</th><th>Name</th><th>Meaning</th></tr><tr><td>0</td><td>Reserved</td><td>Unused.</td></tr><tr><td>1</td><td>Channel Overridden</td><td>The channel is not responding to C-Bus commands because the local override has been activated and C-Bus Priority is disabled.</td></tr><tr><td>2</td><td>Load Incompatible</td><td>The channel has shut down due to the attached load being incompatible with load profile.</td></tr><tr><td>3</td><td>Channel Offline</td><td>Channel is not responding to commands, either due to a loss of power to the channel or a configuration error.</td></tr><tr><td>4</td><td>Mode Error</td><td>The channel has shut down due to the attached load being incompatible with the currently selected mode.</td></tr><tr><td>5</td><td>Over-Temperature Wind-Back</td><td>The channel is operating at a reduced operating point due to excessive temperature.</td></tr><tr><td>6</td><td>Over Current Shutdown</td><td>The channel has shut down due to an extreme current overload.</td></tr><tr><td>7</td><td>Over Temperature Shutdown</td><td>The channel has shut down due to extreme temperature.</td></tr></table> | Bit                        | Name                                                                                                                          | Meaning | 0 | Reserved     | Unused.                                                                                                | 1 | Channel Overridden | The channel is not responding to C-Bus commands because the local override has been activated and C-Bus Priority is disabled. | 2 | Load Incompatible      | The channel has shut down due to the attached load being incompatible with load profile.                 | 3   | Channel Offline | Channel is not responding to commands, either due to a loss of power to the channel or a configuration error. | 4 | Mode Error | The channel has shut down due to the attached load being incompatible with the currently selected mode. | 5 | Over-Temperature Wind-Back | The channel is operating at a reduced operating point due to excessive temperature. | 6 | Over Current Shutdown | The channel has shut down due to an extreme current overload. | 7 | Over Temperature Shutdown | The channel has shut down due to extreme temperature. |
|                                                                                                                                                                                                                               |                     | Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Name                       | Meaning                                                                                                                       |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Reserved                   | Unused.                                                                                                                       |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Channel Overridden         | The channel is not responding to C-Bus commands because the local override has been activated and C-Bus Priority is disabled. |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Load Incompatible          | The channel has shut down due to the attached load being incompatible with load profile.                                      |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Channel Offline            | Channel is not responding to commands, either due to a loss of power to the channel or a configuration error.                 |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mode Error                 | The channel has shut down due to the attached load being incompatible with the currently selected mode.                       |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Over-Temperature Wind-Back | The channel is operating at a reduced operating point due to excessive temperature.                                           |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Over Current Shutdown      | The channel has shut down due to an extreme current overload.                                                                 |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Over Temperature Shutdown  | The channel has shut down due to extreme temperature.                                                                         |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
| <b>NOTE:</b> Error reports are only generated for channels which are in use (have a Group Address assigned).                                                                                                                  |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |                                                                                                                               |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
| 250                                                                                                                                                                                                                           | Power Supply Errors | <table><tr><th>Bit</th><th>Name</th><th>Meaning</th></tr><tr><td>0</td><td>Load Warning</td><td>The load current on the C-Bus Power Supply has exceeded the user-configurable warning threshold level.</td></tr><tr><td>1</td><td>Overload</td><td>The load current on the C-Bus Power Supply has exceeded the rated load current.</td></tr><tr><td>2</td><td>Current Limit Exceeded</td><td>The Power Supply is in current-limit mode, meaning the network communications are likely to be impaired.</td></tr><tr><td>3-7</td><td>Reserved</td><td>Unused.</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Bit                        | Name                                                                                                                          | Meaning | 0 | Load Warning | The load current on the C-Bus Power Supply has exceeded the user-configurable warning threshold level. | 1 | Overload           | The load current on the C-Bus Power Supply has exceeded the rated load current.                                               | 2 | Current Limit Exceeded | The Power Supply is in current-limit mode, meaning the network communications are likely to be impaired. | 3-7 | Reserved        | Unused.                                                                                                       |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Name                       | Meaning                                                                                                                       |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Load Warning               | The load current on the C-Bus Power Supply has exceeded the user-configurable warning threshold level.                        |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Overload                   | The load current on the C-Bus Power Supply has exceeded the rated load current.                                               |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
|                                                                                                                                                                                                                               |                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Current Limit Exceeded     | The Power Supply is in current-limit mode, meaning the network communications are likely to be impaired.                      |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
| 3-7                                                                                                                                                                                                                           | Reserved            | Unused.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                            |                                                                                                                               |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
| The load warning and over-current errors are optional. If the warning threshold parameters are not configured, both these errors are disabled. Once the warning threshold parameters are configured, both errors are enabled. |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |                                                                                                                               |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
| The Current Limit error is always enabled, however due to the nature of the fault, it may only ever appear when most severe errors are enabled, as the network communication may be impaired by this condition.               |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |                                                                                                                               |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |
| <b>NOTE:</b> Power Supply Errors are only generated if the integrated C-Bus Power Supply is enabled.                                                                                                                          |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |                                                                                                                               |         |   |              |                                                                                                        |   |                    |                                                                                                                               |   |                        |                                                                                                          |     |                 |                                                                                                               |   |            |                                                                                                         |   |                            |                                                                                     |   |                       |                                                               |   |                           |                                                       |

| 251 | Aggregated Channel Errors | Same as channel error format<br><br>This channel is the logical OR of the individual channel errors. When used in conjunction with Error Modes 3/4 and a Trigger event for requesting update, this can be useful to minimize Error Reporting traffic. To fetch detailed channel error information, configure the Error Reporting Request Trigger Group parameter, and send a Trigger event using Action Selector 255 to request all channels, or request individual channels by using the Action Selector corresponding to the Error Channel of Interest.                                                                                                                                                                                                                                                                                                                                                |  |  |     |      |         |   |                |                                                                             |   |                       |                                                                                                      |   |                       |                                                                              |   |                        |                                                                               |     |          |         |
|-----|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-----|------|---------|---|----------------|-----------------------------------------------------------------------------|---|-----------------------|------------------------------------------------------------------------------------------------------|---|-----------------------|------------------------------------------------------------------------------|---|------------------------|-------------------------------------------------------------------------------|-----|----------|---------|
| 254 | Unit Errors               | <table><tr><th>Bit</th><th>Name</th><th>Meaning</th></tr><tr><td>0</td><td>Config Invalid</td><td>The unit is not operating correctly because the configuration is not valid.</td></tr><tr><td>1</td><td>Unit Over-Temperature</td><td>The unit's internal operating temperature has exceeded the configured temperature warning threshold.</td></tr><tr><td>2</td><td>C-Bus Voltage Warning</td><td>The C-Bus Voltage has dropped below the user-configurable warning threshold.</td></tr><tr><td>3</td><td>C-Bus Voltage Critical</td><td>The C-Bus Voltage has dropped below the user-configurable critical threshold.</td></tr><tr><td>4–7</td><td>Reserved</td><td>Unused.</td></tr></table><br>The thresholds for the Unit Over-Temperature, and C-Bus Voltage Warning and Critical are user configurable and should be set in accordance with the specific circumstances of the unit in question. |  |  | Bit | Name | Meaning | 0 | Config Invalid | The unit is not operating correctly because the configuration is not valid. | 1 | Unit Over-Temperature | The unit's internal operating temperature has exceeded the configured temperature warning threshold. | 2 | C-Bus Voltage Warning | The C-Bus Voltage has dropped below the user-configurable warning threshold. | 3 | C-Bus Voltage Critical | The C-Bus Voltage has dropped below the user-configurable critical threshold. | 4–7 | Reserved | Unused. |
| Bit | Name                      | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |     |      |         |   |                |                                                                             |   |                       |                                                                                                      |   |                       |                                                                              |   |                        |                                                                               |     |          |         |
| 0   | Config Invalid            | The unit is not operating correctly because the configuration is not valid.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |     |      |         |   |                |                                                                             |   |                       |                                                                                                      |   |                       |                                                                              |   |                        |                                                                               |     |          |         |
| 1   | Unit Over-Temperature     | The unit's internal operating temperature has exceeded the configured temperature warning threshold.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |     |      |         |   |                |                                                                             |   |                       |                                                                                                      |   |                       |                                                                              |   |                        |                                                                               |     |          |         |
| 2   | C-Bus Voltage Warning     | The C-Bus Voltage has dropped below the user-configurable warning threshold.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |     |      |         |   |                |                                                                             |   |                       |                                                                                                      |   |                       |                                                                              |   |                        |                                                                               |     |          |         |
| 3   | C-Bus Voltage Critical    | The C-Bus Voltage has dropped below the user-configurable critical threshold.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |     |      |         |   |                |                                                                             |   |                       |                                                                                                      |   |                       |                                                                              |   |                        |                                                                               |     |          |         |
| 4–7 | Reserved                  | Unused.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |     |      |         |   |                |                                                                             |   |                       |                                                                                                      |   |                       |                                                                              |   |                        |                                                                               |     |          |         |

## Channel Errors

Channel Errors are error conditions relating to each individual channel. For the dimmer units, the following conditions may be reported :

### **Channel Offline**

This indicates that the channel is not operational, either due to loss of power to the channel, the dimmer channel firmware requiring an update, or a configuration error.

### **Channel Overridden**

This indicates that the channel is in a state where it has been overridden by the Local Toggle, but C-Bus Priority has been disabled in its configuration, meaning it will not respond to C-Bus messages indefinitely, until the Local Override is cleared by user input on the unit itself.

### **Load Incompatible**

This indicates that the channel has shut down due to incompatibility of the attached load with the Load Profile selected for the channel.

### **Mode Error**

This indicates that the channel has shut down due to the attached load being incompatible with the currently selected mode. Usually due to detection of an inductive load when the channel is set to TE mode.

### **Over-Temperature Wind-Back**

This indicates that the channel brightness has been reduced due to the channel temperature exceeding 80 °C.

### **Over-Temperature Shutdown**

This indicates that the channel has shut down due to the channel temperature exceeding 100 °C.

### **Over-Current Shutdown**

This indicates that the channel has been shut down due to the current exceeding the over current limit.

## Power Supply Errors

The Power Supply error channel reports on errors related to the inbuilt switchable C-Bus Power Supply. The load current and output voltage of the C-Bus Power Supply are monitored.

Once a C-Bus Network is fully installed, the load on each Power Supply is very stable, however it will be different in different units depending on the distribution of power supplies, the number and distribution of units consuming power, and the cabling between them.

A change in the load on any given C-Bus Power Supply could be a result of additional units being added to the network, or a cable becoming disconnected, or a problem with Power Supply.

### **Load Warning and Overload**

The Load Warning error is optional, and enabled by configuring values for the C-Bus Power Supply Warning Set Threshold and C-Bus Power Supply Warning Clear Threshold. These are settings in mA, corresponding to the load on the Power Supply.

Once these thresholds are configured, both the Load Warning and Overload error monitoring become active. If the load current exceeds the C-Bus Power Supply Warning Set Threshold value the Load Warning Error is raised, if the load also exceeds the rated current of 200 mA, then the Overload error is raised.

For the Overload error to be cleared, the load must drop below 195 mA. For the Load Warning error to be cleared, the load must drop below the C-Bus Power Supply Warning Clear Threshold.

### Current Limit Exceeded

The Current Limit Exceeded error is always enabled, and is raised when the load has exceeded the rated current of 200 mA and the Power Supply output voltage has dropped more than 2 V below the normal level at full load.

This condition indicates a severe overload on the system, possibly due to a short on the C-Bus Network, or a failed Power Supply. As network communication may not be possible under these conditions, this error message may not be observed until the network communication recovers, in which case it will only be reported if Error Mode 2 or 4 is activated.

## Unit Errors

The unit error channel report errors relating to the unit itself. These conditions may indicate an issue with the performance of the unit itself or of the C-Bus Network.

### Config Invalid

This error means the unit is not operational as its stored configuration is incompatible with its firmware version. This can happen if the firmware has been updated with a new major release version, but the unit has not yet been reconfigured to suit using the SpaceLogic C-Bus Commission software.

### Unit Over-Temperature Error

The unit has internal temperature monitoring. This is measuring the temperature at a location on the Power-Supply, and normally measures higher than the ambient temperature of the enclosure. If temperature exceeds the *Unit Over Temperature Set Threshold* (default set to 70 °C), the Unit Over Temperature error condition is set.

Once set, the temperature needs to drop below the *Unit Over Temperature Clear Threshold* to clear the error condition. The thresholds for the Unit Over Temperature error can be changed using the SpaceLogic C-Bus Commission software.

**NOTE:** The Unit Over Temperature error condition does not change the operation of the unit and is provided for information only. It should be noted however that operating at excessive temperature reduces the life of the electronic components.

### C-Bus Voltage Warning/Critical Errors

The unit can monitor the C-Bus Network Voltage and report error conditions based on the measurement.

Once a C-Bus Network is fully installed, the network voltage is very stable, however it will be different at different points in the network depending on the distribution of power supplies, the number and distribution of units consuming power, and the cabling between them. A change in the C-Bus Network voltage could be a result of additional units being added to the network, or a cable becoming disconnected, or a problem with Power Supply.

As what is normal for each unit and installation may be different, each unit can have different thresholds configured to generate error messages if the C-Bus voltage goes outside a desirable window of operation. There are independent error thresholds and conditions for *Warning* and *Critical* levels. These operate independently, with the only constraint being that the threshold values for Warning must be higher than the threshold values for Critical.

**NOTE:** These error conditions do not change the operation of the unit, they are for information only.

# Live Viewing of Monitored Data

The current values of each of the monitored conditions are visible in SpaceLogic C-Bus Commission when connected to the live network.

This is found under the *Status* section at the bottom of the **Property Editor** window.

The screenshot shows the 'Property Editor' window for a C-Bus device. The 'Status' section is expanded, showing two sub-sections: 'Device Status' and 'Channel Status'.

**Device Status**

| Property                          | Value |
|-----------------------------------|-------|
| Hardware Version                  | 1.0   |
| Firmware Version                  | 1.1.0 |
| C-Bus Clock Active                | No    |
| C-Bus Voltage (V)                 | 30.2  |
| Inbuilt C-Bus Power Supply Active | On    |
| Power Supply Load (mA)            | 149   |
| Power Supply Output Voltage (mV)  | 34462 |
| Load Power (mW)                   | 5843  |
| Unit Temperature (°C)             | 51    |

**Channel Status**

| Channel                    | Ch 1 | Ch 2 | Ch 3 | Ch 4 |
|----------------------------|------|------|------|------|
| Load Incompatible          | No   | No   | No   | No   |
| Dimming Mode               | TE   | TE   | TE   | TE   |
| Offline                    | No   | No   | No   | No   |
| Dimming Mode Error         | No   | No   | No   | No   |
| Temperature Wind Back      | No   | No   | No   | No   |
| Temperature Shut Down      | No   | No   | No   | No   |
| Over Current               | No   | No   | No   | No   |
| Operating Temperature (°C) | 42   | 41   | 39   | 40   |
| Mains Frequency (Hz)       | 49.8 | 49.8 | 50.1 | 50.1 |

**Status**

Save to Project

Save Cancel

PROPERTIES DEPLOYMENT QUEUE

## Firmware Update

The C-Bus Digital Dimmers contain multiple microprocessors, all of which can be updated. From time to time, new firmware may be released to enhance performance or add new functionality. The SpaceLogic C-Bus Commission software will notify if a firmware update is required.

Firmware updates are performed using the SpaceLogic C-Bus Commission software, and require a connection to the USB-C connector on the front end of the dimmer.

All required firmware files will be included and authenticated as part of the latest SpaceLogic C-Bus Commission software release.

## Technical Data

| Parameter                           | Description                                                                                                                                                                                |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nominal Mains Voltage and Frequency | 230 Vac $\pm$ 10 %, 50 Hz $\pm$ 3 Hz<br>110 Vac $\pm$ 10 %, 60 Hz $\pm$ 3 Hz                                                                                                               |
| C-Bus Input Operating Voltage       | 20 - 36 Vdc                                                                                                                                                                                |
| * C-Bus Input Current               | 20 mA                                                                                                                                                                                      |
| C-Bus Power Supply (if enabled)     | 200 mA @ 27-35 Vdc                                                                                                                                                                         |
| Load Rating                         | 5508D1D: Up to 1 A per channel, Max 8 A total<br>5504D2D: Up to 2 A per channel, Max 8 A total<br>5504DHD: Up to 5 A per channel, Max 6 A total on any 2 adjacent channels, Max 12 A total |
| Ingress Protection                  | IP20                                                                                                                                                                                       |
| Dimmer Technology                   | Trailing Edge (TE) or Leading Edge (LE) -Selectable per channel                                                                                                                            |
| Compatible Load Types               | Dimmable LED lamps and luminaires, LV incandescent/halogen lamps, ELV halogen lamps with dimmable electronic transformers, ELV halogen lamps with dimmable iron-core transformers          |
| Terminals                           | Accommodates 2 x 2.5 mm <sup>2</sup> or 1 x 4 mm <sup>2</sup> cable<br>Require Philips PH1 or 0.8 x 4.0 flat blade screwdriver<br>Maximum torque 0.6 Nm                                    |
| C-Bus Connection                    | 2 x RJ45 Connectors                                                                                                                                                                        |
| Operating Temperature               | -5 to 55 °C (with derating above 35 °C)                                                                                                                                                    |
| Operating Humidity                  | 10 % to 93 % (non-condensing)                                                                                                                                                              |
| Storage Temperature                 | 20 to 60 °C                                                                                                                                                                                |
| Dimensions                          | 216 mm (W)<br>92 mm (H)<br>53 mm (D)                                                                                                                                                       |

\* 20 mA is consumed from C-Bus input only if the integrated C-Bus Power Supply is disabled. If the integrated C-Bus Power Supply is enabled, 20 mA is consumed internally and 200 mA is available for external consumption.

# Troubleshooting

| Symptom                                                                       | Possible Indication                                                   | Possible Causes                                                                                                                                                                                                                        | Solution                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reduced brightness observed in the dimmer channel outputs                     | <b>Channel Indicator:</b> Slow Flash, Yellow.                         | <ul style="list-style-type: none"> <li>Channel is operating at reduced setting due to over temperature condition.</li> <li>Normally due to excessive load and/or inadequate ventilation and/ or high operating temperature.</li> </ul> | <ul style="list-style-type: none"> <li>Reduce the load on the channel and/or total the load on the unit.</li> <li>Improve enclosure ventilation to reduce the temperature of the enclosure.</li> </ul>                                    |
| Dimmer Channel does not work                                                  | <b>Channel Indicator:</b> Fast Flash, Yellow.                         | <ul style="list-style-type: none"> <li>Channel shutdown due to over temperature condition.</li> <li>Normally due to excessive load and/or inadequate ventilation.</li> </ul>                                                           |                                                                                                                                                                                                                                           |
|                                                                               | <b>Channel Indicator:</b> Fast Flash, Red.                            | <ul style="list-style-type: none"> <li>Channel shutdown due to over current.</li> <li>Normally due to excessive inrush or load/wiring fault (short circuit).</li> </ul>                                                                | Check the wiring, rectify the issue and then turn on the dimmer to restore it to its normal operating state.                                                                                                                              |
| Channel does not respond to Local Override buttons                            | Not applicable                                                        | <ul style="list-style-type: none"> <li>Local Override feature has been disabled by the System Integrator.</li> </ul>                                                                                                                   | <ul style="list-style-type: none"> <li>Re-enable feature using SpaceLogic C-Bus Commission software.</li> </ul>                                                                                                                           |
| Channel in LE Mode does not turn on or turns off immediately after turning on | Channel Indicator: Slow Flash, Red.                                   | <ul style="list-style-type: none"> <li>Load is incompatible with current load profile.</li> <li>Electronic Load.</li> </ul>                                                                                                            | <ol style="list-style-type: none"> <li>Change load profile settings using SpaceLogic C-Bus Commission software.</li> <li>Use the channel button the change to TE mode.</li> </ol>                                                         |
| Channel in TE Mode does not turn on or turns off immediately after turning on | <b>Channel Indicator:</b> Slow Flash, Red.                            | Mode Error (Inductive Load detected).                                                                                                                                                                                                  | Use the channel button to change to LE mode.                                                                                                                                                                                              |
| Lights on channel in TE mode turn on very slowly                              | Channel Indicator: Normal (Yellow)                                    | Channel is in Load Detection Mode (Mode Confirmation process has not been performed).                                                                                                                                                  | Perform Mode Confirmation process.                                                                                                                                                                                                        |
| Unit does not respond to C-Bus commands                                       | <b>C-Bus indicator:</b> Solid Red                                     | No C-Bus clock detected.                                                                                                                                                                                                               | Enable C-Bus system clock generator.                                                                                                                                                                                                      |
|                                                                               | <b>Unit indicator:</b> Solid Red                                      | Stored configuration is not compatible with firmware version Can occur after a major firmware version update.                                                                                                                          | Redeploy the configuration of the unit using the latest version of SpaceLogic C-Bus Commission software.                                                                                                                                  |
|                                                                               | <b>Unit indicator:</b> Blinking<br><b>Channel indicators:</b> All Off | <ol style="list-style-type: none"> <li>Remote Off override is activated intentionally.</li> <li>Remote Off override is activated due to a wiring fault in the C-Bus cable.</li> </ol>                                                  | <ol style="list-style-type: none"> <li>Deactivate Remote Off override.</li> <li>Check C-Bus network wiring for shorts from Remote Off pairs (Brown and Brown/White) to the C-Bus Negative pairs (Orange/White and Blue/White).</li> </ol> |
|                                                                               | <b>Unit indicator:</b> Blinking<br><b>Channel indicators:</b> All On  | <ol style="list-style-type: none"> <li>Remote On override is activated intentionally.</li> <li>Remote On override is activated due to a wiring fault in the C-Bus cable.</li> </ol>                                                    | <ol style="list-style-type: none"> <li>Deactivate Remote On override.</li> <li>Check C-Bus network wiring for shorts from Remote On pairs (Green and Green/White) to the C-Bus Negative pairs (Orange/White and Blue/White).</li> </ol>   |
| Cannot scan the unit on the C-Bus network                                     | Not applicable                                                        | Unit is in bootloader state.                                                                                                                                                                                                           | Cycle the power on the unit to exit bootloader.                                                                                                                                                                                           |
|                                                                               | <b>Unit indicator:</b> Off                                            | No mains Power Supply.                                                                                                                                                                                                                 | Check there is mains power to the unit.                                                                                                                                                                                                   |
|                                                                               | <b>C-Bus indicator:</b> Off                                           | C-Bus cable not connected.                                                                                                                                                                                                             | Check that the C-Bus cable is terminated correctly and connected properly.                                                                                                                                                                |
|                                                                               | <b>C-Bus indicator:</b> Solid Red                                     | No C-Bus clock detected.                                                                                                                                                                                                               | Enable C-Bus system clock generator.                                                                                                                                                                                                      |
| Channel does not respond to C-Bus Commands                                    | <b>Channel indicator:</b> Solid Red                                   | <ol style="list-style-type: none"> <li>Channel Offline due to invalid configuration.</li> <li>Channel offline due to incompatible firmware version.</li> </ol>                                                                         | <ol style="list-style-type: none"> <li>Re-deploy the configuration of the unit using the latest version of SpaceLogic C-Bus Commission software.</li> <li>Perform firmware update using SpaceLogic C-Bus Commission.</li> </ol>           |



| Symptom                                                                             | Possible Indication                           | Possible Causes                                                                                                                   | Solution                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unexpected Indicator Flashing                                                       | <b>Unit indicator:</b> Slow Flash, Red/Yellow | Internal temperature has exceeded over-temperature threshold (Power Supply currently disabled).                                   | Improve enclosure ventilation to reduce ambient temperature.                                                                                                                                                                                                    |
|                                                                                     | <b>Unit indicator:</b> Slow Flash, Red/Green  | Internal temperature has exceeded over-temperature threshold (Power Supply currently enabled).                                    | Improve enclosure ventilation to reduce ambient temperature. Consider disabling the C-Bus Power Supply in this unit and enabling it in a unit where the ambient temperature may be lower (for example a unit lower in the switchboard).                         |
| Cannot change the C-Bus Power Supply state from the front panel                     | Not applicable                                | Front panel Power Supply toggle feature has been disabled by the System Integrator.                                               | Re-enable feature using SpaceLogic C-Bus Commission software.                                                                                                                                                                                                   |
| Cannot enable the C-Bus system clock from the front panel                           | Not applicable                                | Front panel system clock toggle feature has disabled by the System Integrator.                                                    | Re-enable feature using SpaceLogic C-Bus Commission software.                                                                                                                                                                                                   |
| No C-Bus Power                                                                      | <b>Unit indicator:</b> Off                    | No mains .                                                                                                                        | Check mains voltage.                                                                                                                                                                                                                                            |
|                                                                                     | Not applicable                                | C-Bus cable not connected.                                                                                                        | Check C-Bus cable.                                                                                                                                                                                                                                              |
|                                                                                     | Not applicable                                | Jumper link not fitted for 110 V.                                                                                                 | Check the jumper is fitted if using a 110 V supply.                                                                                                                                                                                                             |
| Low C-Bus Voltage                                                                   | <b>C-Bus indicator:</b> Steady Flash          | Overload on the network.                                                                                                          | Enable C-Bus Power Supply in additional units or split network if 2000 mA limit has been reached.                                                                                                                                                               |
|                                                                                     | <b>C-Bus indicator:</b> Short Flash           | <ol style="list-style-type: none"> <li>1. Low mains voltage.</li> <li>2. Voltage drop over long distance of cable run.</li> </ol> | <ol style="list-style-type: none"> <li>1. Advise to check mains voltage.</li> <li>2. Enable C-Bus Power Supply in this unit.</li> </ol>                                                                                                                         |
| The lights connected to the output are flickering                                   | Not applicable                                | The default load profile might not be suitable for this light.                                                                    | Customize the load profile and dimming curve to achieve the desired dimming outcome.                                                                                                                                                                            |
|                                                                                     |                                               | Lights can be non-dimmable light sources.                                                                                         | Ensure the load that is connected to the dimmer is a dimmable load.                                                                                                                                                                                             |
|                                                                                     |                                               | If it occurs at specific times, usually in the evening, there maybe ripple signalling on the grid.                                | Choose a different type of lamp which is less affected by ripple, or install a ripple filter.                                                                                                                                                                   |
| It is difficult to set the brightness level at the bottom or top end of the dimming | Not applicable                                | The default 1:1 dimming curve may not be suitable for the loads profile.                                                          | Customize the load profile and dimming curve to achieve the desired dimming outcome.                                                                                                                                                                            |
| Lights flashes brightly at start up                                                 | Not applicable                                | The default load profile is not suitable for the load or incorrect setting for Kick-start.                                        | <ul style="list-style-type: none"> <li>• Customize the default load profile with minimum and maximum brightness to make the lighting behavior suitable with the load.</li> <li>• Ensure there is no incorrectly configured Kick-Start parameter set.</li> </ul> |
| The lights do not turn on at lower brightness level set by C-Bus                    | Not applicable                                | The default load profile is not suitable for the load.                                                                            | <ul style="list-style-type: none"> <li>• Customize the default load profile with minimum and maximum brightness to make the lighting behavior suitable with the load.</li> <li>• Consider configuring Kick-Start.</li> </ul>                                    |
| It is difficult to set the brightness level at the bottom or top end of the dimming | Not applicable                                | The default 1:1 dimming curve may not be suitable for the loads profile.                                                          | Customize the load profile and dimming curve to achieve the desired dimming outcome.                                                                                                                                                                            |
| Light flashes brightly at start up                                                  | Not applicable                                | The default load profile is not suitable for the load or incorrect setting for Kick-start.                                        | <ul style="list-style-type: none"> <li>• Customize the default load profile with minimum and maximum brightness to make the lighting behavior suitable with the load.</li> <li>• Ensure there is no incorrectly configured Kick-Start parameter set.</li> </ul> |

| Symptom                                                          | Possible Indication | Possible Causes                                        | Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------|---------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The lights do not turn on at lower brightness level set by C-Bus | Not applicable      | The default load profile is not suitable for the load. | <ul style="list-style-type: none"> <li>Customize the default load profile with minimum and maximum brightness to make the lighting behavior suitable with the load.</li> <li>Consider configuring Kick-Start.</li> </ul>                                                                                                                                                                                                                                                                                                                                                               |
| Unit does not work after firmware update                         | Not applicable      | Interruption during firmware update.                   | <ul style="list-style-type: none"> <li>Check the condition of the USB communication cable used between the laptop and the device and ensure good quality cable is used for firmware update.</li> <li>Try a shorter cable.</li> <li>Ensure update process is not interrupted, and wait for software to confirm process is complete before removing the cable.</li> <li>Check the USB cable connection and ensure it is fully inserted in the USB-C connector.</li> <li>If instructed by the software perform an extra-short press the HR reset button for recovery purposes.</li> </ul> |

## Product Environmental Data

Find and download comprehensive environmental data about your products, including RoHS compliance and REACH declarations as well as Product Environmental Profile (PEP), End-of-Life instructions (EOLi) and much more.

<https://www.se.com/myschneider>



## General information about Schneider Environmental Data Program

Click the link below to read about Schneider Electric's Environmental Data Program.

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